CORE MATHEMATICS GRADE 10: VECTORS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.8 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.8: Vectors
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Vector and scalar quantities – Vector notation – Representation of vectors – Equivalent vectors – Addition of vectors – Multiplication of vectors by a scalar – Column vectors – Position vectors – Magnitude of a vector – Midpoint of a vector – Translation vector
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain vector and scalar quantities with illustrations, b) use vector notations in different ways, c) represent vectors geometrically in different situations, d) identify equivalent vectors in different situations, e) add vectors in different situations, f) multiply vectors by scalars, g) determine column vectors in different situations, h) determine position vectors in different situations, i) work out magnitude of a vector in different situations, j) determine the midpoint of a vector in different situations, k) determine translation vector as a transformation, l) appreciate the use of vectors in real-life situations.
Core Competencies	– Learning to Learn: As the learner uses charts and grids to practise addition of vectors, building independent mathematical reasoning and persistence in problem solving.

	– Self-Efficacy: As the learner shares and uses resources from the immediate environment such as plane figures or flip charts to demonstrate equivalent vectors, developing confidence in applying vector concepts to real contexts.
Core Values	– Responsibility: As the learner uses resources from the immediate environment such as plane figures, flip charts and relates equivalent vectors, taking care of shared learning materials and representing data accurately. – Unity: As the learner searches and brainstorms on the meaning of the terms vector and scalar quantities and illustrates giving examples, working collaboratively and valuing each member's contribution to the group.
Science & Engineering Practices	– Developing and Using Models: Learners construct and revise geometric vector diagrams on grids and Cartesian planes to model real-world displacements (e.g., a matatu route from Nairobi's CBD to Westlands), building cumulative representational models across lessons. – Analysing and Interpreting Data: Learners analyse column vector data and position vectors plotted on Cartesian coordinates, interpreting magnitudes and directions to draw conclusions about resultant motion. – Constructing Explanations: Learners use vector addition diagrams and algebraic notation to construct evidence-based explanations for why road sign arrows and displacement cues indicate direction and magnitude simultaneously. – Engaging in Argument from Evidence: Learners compare scalar versus vector descriptions of journeys (e.g., "5 km" vs. "5 km due north from Kisumu pier") and argue from evidence why direction matters in real navigation.
Pertinent & Contemporary Issues (PCIs)	– Education for Sustainable Development – Environmental Education: As learners practise addition of vectors using charts and grids made from resources in the immediate environment, promoting resourcefulness and environmental consciousness. – Road Safety: As the learner discusses the meaning of arrows and lines on roads and road signs in relation to movement on the roads, reinforcing the importance of directional awareness and safe behaviour for pedestrians and drivers.
Career Connections	– Engineering (Civil, Structural, Mechanical): Vectors are fundamental to force analysis, structural load calculations, and motion planning — directly relevant to Kenya's infrastructure development (e.g., SGR railway, LAPSSET corridor). – Navigation and Aviation: Pilots and maritime navigators at Jomo Kenyatta International Airport and Mombasa Port use vector principles to compute resultant displacement accounting for wind and currents. – Physics and Applied Sciences: University-level kinematics, electromagnetism, and fluid mechanics all build directly on vector algebra introduced here. – Computer Graphics and Game Development: Kenyan tech hubs like Nairobi's Silicon Savannah (iHub) use vectors extensively in 2D/3D rendering, animation, and game physics engines. – Surveying and Cartography: Land surveyors working across the Rift Valley and coastal regions apply position vectors and translation vectors to map terrain and demarcate boundaries accurately.
Focus for Lessons	This unit introduces learners to the concept of vectors as mathematical objects that carry both magnitude and direction, contrasting them with scalar quantities encountered in everyday Kenyan contexts such as distances on matatu routes and wind direction across Lake Victoria. Using grids, Cartesian planes, and locally available materials, learners progress from geometric representation through to algebraic operations — addition, scalar multiplication, column vectors, position vectors, magnitude, midpoints, and translation — building a coherent, cumulative understanding. The unit anchors all learning in a real-world phenomenon of directional navigation so that learners appreciate why direction, not just distance, is essential in real-life situations.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely? KICD KEY INQUIRY QUESTIONS: 1. What is the meaning of the terms vector and scalar quantities?

	2. Where are vectors used in real-life situations?
Anchoring Phenomenon	ANCHORING PHENOMENON: Two matatu drivers both travel exactly 8 km from Kencom Bus Stop in Nairobi's CBD, yet one arrives at Westlands and the other arrives at South B — completely different destinations. Students are shown a simplified Nairobi street grid map with both routes marked. Both routes are labelled "8 km" in distance, yet the endpoints are far apart. This is puzzling because learners' prior understanding of distance (a scalar) suggests that travelling the same distance should bring you to equivalent places. The phenomenon exposes the insufficiency of magnitude alone and creates the need for a new mathematical object — one that captures both how far and in which direction — which is the vector. The map also shows road directional arrows and one-way signs, prompting learners to notice that direction is already encoded in everyday road infrastructure, setting up the driving question for the entire unit.
Storyline Thread	Lesson 1: ANCHOR + PREDICT — Launch the anchoring phenomenon (two 8 km matatu routes, different destinations). Learners predict in writing why the same distance produces different endpoints. Introduce the terms vector and scalar; learners classify everyday quantities (temperature in Nairobi, wind speed at Kisumu airport, weight of a bag of maize, displacement of a ferry). Begin the Driving Question Board (DQB). Lesson 2: OBSERVE + REPRESENT — Learners use road-map grids and arrow-headed line segments to represent vectors geometrically. Introduce vector notation (bold, underline, or component form). Learners draw vectors on plain paper and squared grids, labelling magnitude and direction. Identify equivalent vectors (same magnitude and direction regardless of position) using cut-outs on a Cartesian grid. Lesson 3: EXPLAIN + ADD VECTORS — Learners investigate vector addition using the triangle law and parallelogram law on grids. Guided activity: map out a two-leg matatu journey (Kencom → University of Nairobi → GPO) as two vectors and find the resultant (single displacement vector). Practise addition of column vectors algebraically. Connect back to phenomenon: the resultant explains the final destination. Lesson 4: OBSERVE + SCALAR MULTIPLICATION — Learners explore what happens when a vector is multiplied by a positive scalar, a negative scalar, and a fraction using grid diagrams. Context: if a boda-boda rider covers vector **d** in one trip, what does 3**d** represent? Learners draw scaled vectors and identify that scalar multiplication changes magnitude (and reverses direction for negatives) but not the line of action. Subtraction of vectors introduced as adding the negative. Lesson 5: EXPLAIN + COLUMN VECTORS & POSITION VECTORS — Learners express vectors in column form on a Cartesian plane, linking geometric arrows to algebraic notation. Introduce position vectors relative to the origin O; learners plot landmarks on a Nairobi coordinate grid (e.g., O = Kencom, A = Kenyatta International Conference Centre, B = Uhuru Park) and express $\vec{OA}$ and $\vec{OB}$ as column vectors. Derive $\vec{AB} = \vec{OB} - \vec{OA}$. Update DQB with new understanding. Lesson 6: OBSERVE + MAGNITUDE & MIDPOINT — Learners apply Pythagoras' theorem to derive the magnitude formula |**v**| = $\sqrt{x^2 + y^2}$. Contextualised problems: find the straight-line distance (magnitude) of a displacement from Kisumu pier to a fishing ground given a column vector. Introduce midpoint of a vector: if a relay runner runs from point A to point B along a vector, where is the halfway mark? Practise with Kenyan athletics track scenarios. Lesson 7: EXPLAIN + TRANSLATION VECTOR AS TRANSFORMATION — Learners connect vectors to transformation geometry: a translation moves every point of an object by the same vector. Learners perform translations of simple shapes on a Cartesian plane using a given translation vector, and reverse-engineer the translation vector from an object and its image. Link to road safety context: road arrows as translation instructions. Learners revise and extend their cumulative geometric model on the DQB.

	Lesson 8: MODEL BUILD + CONSOLIDATION & APPLICATION — Learners synthesise all concepts in a group problem-solving task: given a sketch map of a Kenyan town (e.g., Nakuru), describe a multi-leg journey using vectors, compute the resultant displacement, find its magnitude, identify the midpoint of the journey, and represent the translation that maps the start point to the endpoint. Groups present their vector models. Final DQB review: learners answer the driving question in writing, explaining why two equal-distance journeys produce different destinations and how vectors resolve this puzzle. Formative assessment via class activity.
Supporting Phenomena	– A football kicked with the same force (same magnitude) in two different directions during a match at Nyayo National Stadium travels to completely different positions on the pitch — illustrating that magnitude alone does not determine outcome; direction is equally critical. – A fishing boat leaving Kisumu pier on Lake Victoria travels for 2 hours at the same speed on two separate days, but ends up at different landing sites because the wind direction changed — showing how vector quantities (velocity with direction) govern real displacement. – A porter carrying luggage at Mombasa's Likoni Ferry crossing walks the same number of steps each day yet boards different ferries depending on which direction they walk along the jetty — a relatable local context for displacement vs. distance. – Arrows and lane markings painted on Kenyan roads (e.g., the Thika Superhighway) encode direction explicitly; removing the arrowhead leaves only a line segment (scalar length) and makes the sign ambiguous — visually demonstrating the difference between a scalar segment and a vector.

LESSON 1 (80 minutes): Why Does 8 km Take You to Westlands or South B? Launching the Matatu Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon and establish the cognitive need for vectors by exposing the insufficiency of distance (scalar) alone in describing a journey, and to introduce the foundational vocabulary of vector and scalar quantities.
Knowledge	Learners will know: (a) the meaning of vector and scalar quantities with illustrations; (b) that a vector requires both magnitude and direction while a scalar requires magnitude only; (c) everyday examples of vector and scalar quantities in Kenyan contexts.
Skills	Learners will be able to: (a) classify everyday quantities as either vectors or scalars and justify their classification; (b) use vector notations to label directed quantities; (c) pose and record investigative questions about vectors on the Driving Question Board.
Attitudes	Learners will appreciate the relevance of vectors in explaining real-life navigation puzzles such as the matatu routes from Nairobi's CBD, and will value precision in mathematical description — understanding that magnitude alone is insufficient to fully describe movement.
Key Inquiry Question	1. What is the meaning of the terms vector and scalar quantities? 2. Where are vectors used in real-life situations?
Purpose in Storyline	This is the ANCHOR + PREDICT lesson. It launches the anchoring phenomenon — two matatu routes of exactly 8 km from Kencom Bus Stop leading to completely different destinations (Westlands and South B) — to create a productive cognitive conflict. Learners discover that distance alone (a scalar) cannot explain the puzzle, generating the need for a new mathematical object: the vector. The lesson opens the Driving Question Board (DQB), which will grow throughout the unit.
Safety Notes	No physical hazards. Ensure the Nairobi street grid map printout or projected image is clear and legible. If learners are asked to stand and move to simulate journeys, ensure adequate floor space and remind learners to be respectful of peers' physical space.

B. LESSON OVERVIEW
Lesson 1 opens the entire Vectors unit by presenting the anchoring phenomenon: a simplified Nairobi street grid map showing two matatu routes from Kencom Bus Stop, both labelled 8 km, yet ending at Westlands (northwest) and South B (southeast) respectively. This visible contradiction between equal distances and completely different destinations drives a whole-class prediction activity, surfaces prior knowledge of distance and direction, and creates an authentic need for the concept of a vector. The teacher then introduces the terms 'vector' and 'scalar' with Kenyan-context examples, and learners classify a set of everyday quantities. The lesson closes by opening the class Driving Question Board (DQB) where learners post their burning questions for the unit. No computation is required in this lesson; the focus is entirely on conceptual anchoring, vocabulary introduction, and curiosity generation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Multiplying Matrices and Scalars -]	Learners are shown the Nairobi street grid map (projected or on a printed A3 handout) displaying two matatu routes from Kencom Bus Stop: Route A heading northwest	SETUP (before class): Project or print the simplified Nairobi street grid map. Mark Kencom Bus Stop clearly at the centre. Draw Route A (northwest arrow, labelled '8 km →	PREDICT-BEFORE-INSTRUCTION (Elicitation of Prior Knowledge): By writing predictions before any teaching, learners activate and make visible their	Observable indicator: Learners produce a written prediction of at least two sentences in their exercise books. The teacher scans books during circulation — look for

[Overview](#) Source: CK-12 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	to Westlands (8 km) and Route B heading southeast to South B (8 km). Both routes are clearly labelled '8 km' with directional arrows. Learners individually read the map silently for 2 minutes, then write a response in their exercise books to the prompt: 'Both matatus travel exactly 8 km from Kencom. Why do they end up at completely different places? What information is missing from the label '8 km'?' Learners write independently for 5 minutes without discussion. After writing, learners share predictions with a partner (Think-Pair) for 2 minutes. Three to four pairs are cold-called to share their predictions with the class. The teacher records key prediction words on the board without evaluating them (e.g., 'direction', 'road', 'route', 'north', 'turning').	Westlands') and Route B (southeast arrow, labelled '8 km → South B'). Include one-way road signs and directional arrows on a few roads to prime direction thinking. LAUNCH (5 min): Point to the map and say: 'Both of these matatu drivers leave Kencom Bus Stop at exactly the same time. Both drive exactly 8 km. But look — one arrives at Westlands, the other at South B. That makes no sense to me. I thought the same distance means the same place.' Pause. Allow 10 seconds of silence. Then say: 'Open your exercise books. Write the date and the heading: PREDICTION. Answer this question: What information is missing from the label 8 km that would explain how two journeys of the same distance reach completely different places?' INDIVIDUAL WRITE (5 min): Circulate silently. Do not answer questions or affirm any prediction. Observe what vocabulary learners use. Note learners who write 'direction' — they will be useful for discussion. THINK-PAIR (2 min): Say 'Now share your prediction with your partner. Do not change your own writing — just listen and compare.' COLD-CALL SHARE (4 min): Cold-call exactly 4 pairs. Say: 'I am going to hear from four pairs. Tell me your prediction — do not worry if you are wrong, we are just predicting.' After each response, write the key word from their prediction on the board with no judgment. Use: 'Interesting. What made you think that?' Wait 10 seconds after each response before calling the next pair.	existing understanding of distance and direction. Recording key prediction words on the board without evaluation creates a shared reference point the class can return to throughout the lesson and unit, building metacognitive awareness of how their thinking changes.	three categories: (1) Learners who mention 'direction' or equivalent (ready for extension); (2) Learners who mention 'road' or 'route' but not direction (productive partial understanding); (3) Learners who focus only on speed or time (misconception to address — gently note that the problem says both travel 8 km total distance, not time). Teacher mentally notes names in each category for targeted questioning in later phases.
Observe Phase	Learners engage in two structured observation activities. ACTIVITY A	ACTIVITY A SETUP: Write the three observation questions on the	STRUCTURED OBSERVATION WITH GUIDED QUESTIONS: The	Observable indicator: Each group's shared paper contains responses

📹 **VIDEO:** [Multiplying Matrices and Scalars - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	— Map Analysis (10 min): Working in groups of four, learners receive (or examine the projected) Nairobi street grid map and answer three guided observation questions written on the board: (1) 'What is the same about Route A and Route B?' (2) 'What is different about Route A and Route B?' (3) 'Look at the road signs on the map — what information do the arrows on the roads give a driver that the number 8 km does not give?' Groups record answers on a shared piece of paper. ACTIVITY B — Video/Demonstration Clip (8 min): Teacher plays a 3-minute video clip (or conducts a live demonstration using a metre stick and compass rose drawn on the floor) showing two learners walking from the same starting point: one walks 8 steps north, one walks 8 steps east — both take 8 steps but end far apart. After the clip/demonstration, learners individually write one sentence: 'The video shows that to describe a journey completely, you need to know ________ and ________.' Learners share sentence completions with their group before a whole-class share-out.	board before displaying the map. Say: 'In your groups of four, study this map carefully for 10 minutes. Answer all three questions on the paper provided. Every person must write.' Circulate to each group. Targeted prompt for groups stuck on Q3: 'Look at the one-way signs. If I told you to drive 8 km from Kencom, would you know which one-way street to take? What extra information would you need?' Wait 10 seconds after each prompt before speaking again. ACTIVITY B TRANSITION: Say: 'Now I want you to watch/observe carefully. I am not going to explain what you see. You will explain it.' Play the video or conduct the floor demonstration. After the clip, say: 'Complete this sentence in your book: To describe a journey completely, you need to know ________ and ________.' Give 2 minutes of silent individual writing. SHARE-OUT: Cold-call 3 learners. After the third response, say: 'We are hearing the same two words coming up — magnitude AND direction. Hold on to those words. We are going to give them a mathematical name in the next phase.'	three observation questions scaffold learners from noticing (Q1 & Q2) to explaining (Q3), progressively moving from perceptual observation to conceptual inference. The sentence-completion frame ('you need to know ___ and ___') gives all learners a linguistic scaffold to articulate the core insight — that two pieces of information are needed — before formal vocabulary is introduced.	to all three questions. The teacher specifically checks that Q3 responses reference directional information from road signs — this is the key conceptual bridge. The sentence-completion responses should contain the words 'direction' and 'distance/magnitude/how far' (or equivalent). If a group's Q3 answer only references 'road name' without any directional language, the teacher returns to that group with the prompt: 'If I said turn onto Uhuru Highway — is that enough? Which way do I go on Uhuru Highway?'
Explain Phase 📹 **VIDEO:** [Multiplying Matrices and Scalars - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)] Source: CK-12	The teacher formally introduces the terms VECTOR and SCALAR with definitions, and learners immediately apply the definitions to classify a set of eight Kenyan-context quantities. DEFINITION PHASE (8 min): The teacher writes two definitions on the board and learners copy them into their books: 'A SCALAR quantity is described completely by its magnitude (size) only.' 'A VECTOR quantity is described completely only when both its magnitude AND	DEFINITION INTRODUCTION (8 min): Write SCALAR and VECTOR as large headings on the board. Say: 'We now have a name for the two things a journey needs. Let me introduce them.' Write both definitions. Then say: 'Back to our map. Is 8 km a scalar or a vector?' Wait 10 seconds. Cold-call one learner. 'Correct — 8 km is a SCALAR. It has size only. Now what about: 8 km northwest from Kencom towards Westlands — scalar or vector?' Wait 10 seconds.	CONCEPT-EXAMPLE-APPLICATION (Direct Instruction to Structured Practice): The teacher introduces the formal definition, immediately anchors it to the phenomenon (the matatu map), then transfers ownership to learners through the classification activity. The eight Kenyan-context examples are deliberately mixed in difficulty — temperature and time are unambiguous scalars; wind speed and velocity require careful reading of the quantity description	Observable indicator: Learners correctly classify at least 6 of 8 quantities with a written reason. The teacher specifically watches for: (a) Learners who classify 'wind speed at 15 km/h' (item 2) as scalar — they are reading magnitude only and missing the directional phrase 'blowing from Lake Victoria towards the east'; this reveals that they are not reading the full quantity description, a critical habit of mind for vectors. (b) Learners who

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Driving Question Board (DQB) Creation 📹 **VIDEO:**	The class collectively opens the Driving Question Board for the entire Vectors unit. Each learner writes one question they still have — or a new question the lesson	DQB SETUP (before this phase): Prepare a dedicated corner of the classroom wall or a section of the whiteboard labelled 'OUR DRIVING QUESTION BOARD —	DRIVING QUESTION BOARD (DQB) — Collaborative Question Mapping: The DQB is a publicly visible record of the class's collective curiosity. By categorising	Observable indicator: Every learner posts at least one question. The teacher scans questions for quality: (a) Factual recall questions ('What is a

Multiplying Matrices and Scalars - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	has raised — on a sticky note or a torn piece of paper. Questions are grouped by the teacher on a dedicated section of the classroom wall or board under three prompt headers: (A) 'Questions about what vectors ARE'; (B) 'Questions about how vectors WORK mathematically'; (C) 'Questions about where vectors are USED in Kenya and the world.' The class reads all posted questions aloud (teacher reads or nominates readers). The teacher annotates two or three questions with a star (★) to signal 'we will answer this in an upcoming lesson.' The driving question itself is written in large print at the top of the DQB and remains visible for the entire unit.	VECTORS.' Write at the top in large print: 'WHY do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?' Below this, draw three columns with headers (A), (B), (C) as described. INDIVIDUAL QUESTION WRITING (3 min): Distribute sticky notes or small paper slips. Say: 'You have spent this lesson thinking about vectors and scalars. What questions are still in your head? What do you want to know? Write ONE question — it can be about anything connected to vectors. Do not worry if it sounds basic. Every question goes up.' Allow 3 minutes of silent writing. POSTING (3 min): Invite learners to come up in rows and post their questions under the column that fits best. If unsure, teacher helps them decide. READ-ALOUD (2 min): Teacher reads 5–6 questions aloud, selecting a variety from each column. Say: 'Listen to what your classmates are wondering. These are ALL good questions.' STAR ANNOTATION (1 min): Teacher places a star (★) next to 2–3 questions that align closely with upcoming lessons (e.g., 'How do you add two vectors together?', 'Can you show vectors on a graph?'). Say: 'These starred questions — we will answer them by the end of this unit. Watch this board. Every lesson, we will come back and see which questions we can now answer.'	questions into 'what it is', 'how it works', and 'where it's used', learners see the full scope of the unit through their own questions. This strategy builds metacognitive ownership — learners are not passively receiving a syllabus, they are co-authoring the inquiry. Returning to the DQB in future lessons makes learning progress visible and reinforces the storyline thread.	vector?') suggest the learner needs more anchoring to the phenomenon — teacher can verbally prompt: 'You heard the definition today — what about vectors still puzzles you from the matatu story?'; (b) Procedural questions ('How do you add vectors?') signal readiness for computation in upcoming lessons; (c) Application questions ('Do pilots use vectors when flying from Nairobi to Kisumu?') signal strong transfer thinking — celebrate these publicly. The teacher photographs the DQB at the end of the lesson for their own records and lesson-to-lesson continuity tracking.
Model Building	In this opening lesson, the model-building phase is an INITIAL	INITIAL MODEL INTRODUCTION (3 min): Say: 'In mathematics, a	INITIAL MODEL + CLAIM-EVIDENCE-REASONING (CER):	Observable indicator: Learners produce a diagram with two

Phase 📹 **VIDEO:** [Multiplying Matrices and Scalars - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	MODEL — learners draw their first, rough representation of the phenomenon before formal vector notation is taught. Each learner draws the two matatu routes on a blank outline of a simple grid (a 4×4 grid with Kencom marked at the centre) and uses arrows to show Route A to Westlands and Route B to South B. Learners label their arrows with: (a) the distance (8 km) and (b) their best attempt at a direction (e.g., 'NW' or 'towards Westlands'). Below their diagram, learners complete a CLAIM-EVIDENCE-REASONING frame: 'CLAIM: Distance alone is / is not enough to describe a journey completely because…'; 'EVIDENCE: In the matatu example, …'; 'REASONING: This means that a complete mathematical description of a journey must include…'. Learners keep this initial model — it will be revised in every subsequent lesson as new vector concepts are learned.	model is a way of representing a situation so we can analyse it. Every lesson in this unit, you will update your model of the matatu journey as you learn more. Today, you will make your FIRST model — it will be rough and incomplete, and that is completely fine.' Distribute the 4×4 grid template (or ask learners to draw a simple grid in their books). Say: 'Mark Kencom at the centre. Draw Route A going northwest to Westlands with an arrow. Draw Route B going southeast to South B with an arrow. Label each arrow with 8 km and the direction as best you can.' Give 3 minutes. CLAIM-EVIDENCE-REASONING FRAME (5 min): Write the CER frame on the board. Say: 'Below your diagram, complete these three sentences. Use the map, the video/demonstration we saw, and the definitions you wrote today as your evidence. You have 5 minutes — work silently and independently.' Circulate. Prompt any learner who writes 'distance alone is enough': 'Look at your two arrows — do they go to the same place? So is distance alone enough?' After 5 minutes: cold-call 2 learners to read their full CER. After each, say: 'What did others write that is similar or different? Thumbs up if your reasoning matches.' CLOSE THE LESSON (2 min): Say: 'Keep this model safe — it is the beginning of something. By the end of this unit, you will come back to it and add vector notation, column vectors, magnitudes, and more. Today, the most important thing you leave with is this: 8 km is only half the story. A matatu journey is a vector — it needs direction AND distance.	The initial model externalises each learner's current mental representation of the phenomenon, making thinking visible for both the learner and the teacher. The CER frame is a structured argumentation scaffold (aligned to NGSS Science and Engineering Practice 7 — Engaging in Argument from Evidence) that requires learners to connect their diagram (evidence) to the conceptual claim (distance is insufficient) with explicit reasoning. This establishes the habit of justifying mathematical claims with evidence — a core competency across the CBC Senior School curriculum.	directional arrows from Kencom and a complete CER statement of at least three sentences. The teacher checks: (a) That arrows point in DIFFERENT directions (not just different lengths) — if both arrows point the same way, the learner has not yet grasped the directional core of the phenomenon; (b) That the CLAIM sentence explicitly states 'not enough' — if a learner writes 'distance alone IS enough,' flag for individual follow-up; (c) That the REASONING sentence includes the word 'direction' or equivalent. These initial models are collected or photographed at the end of the lesson as a formative baseline data point for the entire unit.

		That is what we are going to master.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) During the Predict phase, which prediction words appeared most frequently on the board? Did any learners spontaneously use the word 'vector' or 'displacement' — and if so, what does this reveal about their prior exposure? (2) In the classification activity, which of the eight quantities caused the most disagreement? Was it the wind speed item (scalar vs. vector depending on whether direction is stated), and did learners understand that the same physical phenomenon can be described as either depending on what information is given? (3) Looking at the DQB questions posted, are there questions that suggest learners are connecting vectors to subjects beyond mathematics (e.g., Physics, Geography, or Aviation)? How might you leverage these cross-curricular connections in upcoming lessons? (4) In the initial model CER frames, how many learners correctly argued that distance is insufficient? What percentage wrote 'direction' in their reasoning? This baseline percentage is your starting point for tracking conceptual growth across the unit. (5) Did the matatu phenomenon feel authentic and puzzling to learners, or did it feel contrived? If learners live near Nairobi, they may have strong prior knowledge of the routes — use this as an asset. If they are from other counties, consider whether a local equivalent (e.g., two routes from Kisumu's Kondele market, or from Nakuru town centre) would deepen the sense of authenticity for your specific class context.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Two matatu routes from Kencom Bus Stop in Nairobi's CBD, both labelled 8 km, arriving at completely different destinations: Westlands (northwest) and South B (southeast). Road directional arrows and one-way signs encode direction in everyday infrastructure. A demonstration/video showed two people walking 8 steps in different directions and ending far apart.
What did I learn?	A SCALAR quantity is described completely by magnitude (size) only — examples: temperature in Nairobi (22°C), time to cook ugali (20 min), distance run by Eliud Kipchoge (32 km). A VECTOR quantity requires both magnitude AND direction to be described completely — examples: displacement of the Likoni Ferry (mainland to island), velocity of a bodaboda heading south on Ngong Road, wind blowing from Lake Victoria eastward toward Kisumu Airport. The label '8 km' on a matatu route is a scalar — it tells us how far but not which way. A complete journey description is a vector.
How does this explain the phenomenon?	The two matatus ended up at different destinations because 8 km describes only the magnitude (scalar) of the journey, not its direction. To distinguish Route A (8 km northwest → Westlands) from Route B (8 km southeast → South B), we need a mathematical object that carries BOTH magnitude AND direction — this object is called a VECTOR. Direction is already encoded in everyday road infrastructure (one-way signs, directional arrows), which means vectors are not abstract — they are built into the world around us in Nairobi's streets.

LESSON 2 (40 minutes): Observe + Represent — Drawing the Matatu Journey as an Arrow	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners move from the language of vectors (Lesson 1) to the geometry of vectors — representing the two matatu journeys as directed arrow-headed line segments on grids, introducing formal notation, and discovering that vectors in different positions can be identical if they share the same magnitude and direction.
Knowledge	– A vector is represented geometrically as a directed line segment (an arrow) whose length encodes magnitude and whose arrowhead encodes direction. – Vectors can be written using bold type (v), underline notation (<u>v</u>), component/column form, or letter-pair notation (AB with arrow overhead). – Two vectors are equivalent (equal) if and only if they have the same magnitude AND the same direction, regardless of where they are drawn on the plane.
Skills	– Draw vectors accurately on plain paper and squared/Cartesian grids, labelling magnitude and direction. – Identify and justify equivalent vectors using cut-outs and grid inspection. – Translate between geometric arrow representation and symbolic vector notation.
Attitudes	– Appreciate that precise mathematical notation eliminates ambiguity — just as road directional signs remove confusion for matatu drivers. – Show curiosity in exploring how the same mathematical object (a vector) can appear in many positions yet remain equivalent.
Key Inquiry Question	How do we represent a vector quantity — like a matatu journey — completely and precisely using geometric and symbolic tools?
Purpose in Storyline	Lesson 1 established WHY magnitude alone is insufficient. Lesson 2 provides the first mathematical tool to fix this: the directed line segment. Learners now represent both matatu journeys (Kencom → Westlands and Kencom → South B) as arrows on a Nairobi grid, see concretely that different arrows encode different journeys, and begin to develop the notation they will use for all subsequent vector operations in the unit.
Safety Notes	No specific hazards. Ensure scissors used for vector cut-outs are handled safely and returned after activity.

B. LESSON OVERVIEW

This lesson is the OBSERVE + REPRESENT phase of the eight-lesson evidence-gathering sequence for Sub-Strand 2.8 Vectors. It takes the conceptual foundation laid in Lesson 1 — that a journey requires both distance and direction — and gives learners the geometric and symbolic language to express that idea precisely. The anchoring phenomenon remains central: two matatu routes both labelled '8 km' from Kencom Bus Stop in Nairobi's CBD produce completely different destinations. In this lesson, learners discover that if they draw each journey as an arrow on a grid map, the two arrows are visibly, undeniably different objects — different direction, different endpoint — even though both have the same length. This makes the insufficiency of scalar distance physically visible on paper.

The lesson progresses through three interconnected activities. First, learners predict what the two arrows would look like before drawing them, surfacing any residual misconceptions about direction. Second, they observe and draw: using a simplified Nairobi street-grid worksheet and squared paper, they construct the two matatu-journey vectors as arrow-headed line segments, label them with magnitude (8 km) and a compass direction, and experiment with multiple representations (bold notation, underline, column form, and AB-arrow notation). Third, they investigate equivalent vectors using paper cut-outs of arrows on a Cartesian grid, discovering that an arrow translated to a new position but keeping its direction and length is still the same vector — a crucial concept for later lessons on vector addition and position vectors.

Throughout, the teacher connects every activity back to the driving question: 'What mathematical tool do we need to describe a journey completely?' By the end of the lesson, learners can answer that question with confidence — a directed line segment, described by both magnitude and direction — and they begin to update the class Driving Question Board with new evidence. The Model Building phase asks learners to revise their Lesson 1 sketch-model of the phenomenon, replacing vague 'distance + direction' labels with formal arrow notation, bringing their model one step closer to the full vector toolkit they will build across the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Reduction of alkynes Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two unlabelled straight-line segments on the board — both 8 cm long but drawn in different directions — and are asked to write in their exercise books: 'Could these two lines represent the same journey? What information is missing from a plain straight line that would make it represent a matatu route fully?' They share predictions with a seat-partner for 90 seconds before the teacher takes responses. Learners also sketch, without guidance, what they think an arrow representing 'travel 8 km north-west from Kencom to Westlands' would look like, annotating their sketch with any labels they think are needed.	Display two equal-length line segments (no arrowheads, no labels) drawn in different orientations on the board or chart paper. Say: 'Look at these two lines. Both are exactly 8 centimetres long. In your book, write your answer to this question — could these two lines be describing the same matatu journey? What is missing?' Allow 10 seconds of silent thinking (WAIT TIME 1). Then say: 'Now sketch what you think a proper arrow-representation of the Kencom-to-Westlands route would look like — label it with anything you feel is important.' Allow 2 minutes of individual work. Cold-call 3 learners to share sketches (choose one who drew only a line, one who added an arrowhead, one who added a label). For each, ask: 'What does your drawing tell us — and what does it still not tell us?' Do NOT correct yet; capture responses in two columns on the board: 'What the drawing shows' | 'What is still missing'.	Elicit-and-Press Prediction: By asking learners to predict before instruction, the teacher surfaces prior knowledge and common incomplete conceptions (e.g., 'a line is enough'). Capturing 'what is missing' in two board columns creates a public record of gaps that the Observe phase will then fill, giving learners a felt need for the new tool.	Teacher scans sketches during the 2-minute drawing period. Look for: (1) Whether learners include an arrowhead — absence indicates they have not yet connected direction to representation. (2) Whether learners label magnitude (8 km) — absence indicates the scalar–vector link from Lesson 1 needs reinforcing. (3) Whether any learner attempts a compass bearing or grid direction label — this indicates readiness for column-vector notation. Use cold-call responses to gauge class-wide readiness before proceeding.
Observe Phase 📹 **VIDEO:** Reduction of alkynes	Learners receive a simplified Nairobi grid-map worksheet showing Kencom Bus Stop at the origin, with a square grid overlaid (each square = 1 km). Two routes	Distribute the Nairobi grid-map worksheets and squared paper. Say: 'Your task is to represent the two matatu journeys as mathematical arrows — directed	Representation Mapping: Learners move between three representations simultaneously — the real Nairobi map (contextual), the geometric arrow on grid paper	Teacher looks for: (1) Correct arrowhead placement — at the destination end (HEAD), not the starting point. A common error is arrowhead at the tail. (2) At least

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	are pre-marked with dashed lines: Route A heading north-west toward Westlands (8 km) and Route B heading south toward South B (8 km). Learners first trace each route as a directed arrow-headed line segment in pencil, then redraw both vectors on a fresh square-grid page, labelling each with: (a) its magnitude (8 km or the equivalent grid length), (b) a compass direction or bearing, and (c) its name in at least TWO forms of vector notation — e.g., bold **a**, underline a, column vector form, and/or the AB-arrow notation. They then receive pre-cut arrow cut-outs of equal length and are asked to place them on a Cartesian grid in three different positions while keeping direction constant, noting what changes and what stays the same.	line segments. Draw each arrow, then label it using the vector notations I am about to show you.' Model ONE example on the board: draw a vector from point K (Kencom) to point W (Westlands) — say: 'I place my pencil at K — that is the TAIL of the vector. I draw the line toward Westlands, and I put the arrowhead at W — that is the HEAD. The arrowhead tells us direction; the length tells us magnitude. Now I label it.' Write on the board: bold **v**, v (underline), KW with an overhead arrow, and the column vector (−5, 6) as an approximation (explain units). Allow 8 minutes of independent drawing. Circulate — every 2 minutes make one targeted stop: (Stop 1) check arrowhead direction is correct; (Stop 2) check at least two notation forms are used; (Stop 3) check the cut-out placement activity is underway. After 8 minutes, cold-call 2 learners to hold up their grid drawings. Ask the class: 'Are these two vectors — the Westlands arrow and the South B arrow — the same vector or different vectors? Turn and talk for 30 seconds.' (WAIT TIME 2 — 30 seconds pair talk.) Cold-call 3 pairs for responses. Use probing question: 'What exactly makes them different — the length, the direction, or both?'	(visual-spatial), and symbolic notation (abstract). This multi-representational approach (consistent with NGSS Science and Engineering Practice 2: Developing and Using Models) anchors the abstract notation in the concrete phenomenon, making it meaningful rather than arbitrary. The cut-out activity adds a kinaesthetic layer that prepares learners for the concept of equivalent/free vectors.	two notation forms correctly used. (3) During the cut-out activity — can learners articulate that moving the arrow to a new grid position does NOT change the vector, only its location? Listen for language like 'same direction, same length' as evidence of conceptual understanding. Exit signal: thumbs-up if you used at least two notation forms; thumbs-sideways if you only used one; thumbs-down if you are unsure — use this to form pairs for the Explain phase.
Explain Phase 📹 **VIDEO:** Reduction of alkynes Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners work in pairs to complete a structured 'Notation Translation Table' — a table with four rows, each giving a vector in ONE notation form and asking learners to express it in the other three. Example row: given the column vector (3, −4), write it as bold notation, underline notation, and describe the geometric arrow	Say: 'You have seen two ways to write and draw vectors. Now work with your partner to complete the Notation Translation Table. You have 6 minutes.' Allow 6 minutes of pair work. Circulate and listen for: confusion between bold and underline (both acceptable in Kenyan Senior School); errors in reading column vector	Notation Translation Table (Structured Sensemaking): Forcing learners to translate between four representations within a single table makes explicit that these are all descriptions of the same mathematical object. This strategy, aligned with NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information),	Collect one Notation Translation Table per pair at the end of the phase (or photograph if paper is scarce). Check for: (1) Correct component ordering in column vectors. (2) Correct use of at least two symbolic notations. (3) Correct identification of equivalent vectors with a clear magnitude-and-direction justification — not just

📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	(direction: 3 units right, 4 units down; magnitude: 5 units by Pythagoras — a preview of magnitude calculation in a later lesson). A second task asks: 'On the Cartesian grid below, vectors P, Q, and R are drawn. Which are equivalent to the Westlands vector (−5, 6)? Circle them and write one sentence explaining why.' This directly links back to the cut-out activity and the phenomenon.	components (x-component first, y-component second). After 6 minutes, pause the class. Say: 'Let us look at the equivalent-vectors task together. I am going to show three arrows on the board — P, Q, and R. I want three volunteers to come and write the column vector for each.' Cold-call 3 learners (aim for gender balance). After each learner writes a column vector, ask the class: 'Is this vector equivalent to our Westlands vector? Give me a thumbs-up or thumbs-down.' (WAIT TIME: 10 seconds per vector.) After all three, say: 'So equivalent vectors must have IDENTICAL column vectors — same x-component, same y-component. This means same magnitude AND same direction. Where those arrows sit on the grid does not matter.' Connect back to phenomenon: 'If I told a matatu driver — your journey is vector (−5, 6) — he knows exactly how far and which way to go, no matter where he starts. This is why we need vector notation, not just the number 8.'	builds representational fluency — the ability to move fluidly between notations — which is essential for all subsequent vector work. The equivalent-vectors task then applies this fluency to a decision-making context rooted in the phenomenon.	'they look the same.' Flag pairs who justify equivalence by visual appearance alone for targeted questioning in the next phase.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Reduction of alkynes] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES	Learners write one new 'We now know…' sticky note and one new 'We still wonder…' sticky note for the class Driving Question Board. The 'We now know' note must contain a geometric or notational claim (e.g., 'A vector is an arrow with a tail and a head; we write it as bold v or as a column (x, y)'). The 'We still wonder' note should push the inquiry forward (e.g., 'Can we add two matatu journeys together as vectors?' or 'How do we calculate the exact length of a vector from its column form?'). Learners post their sticky notes in the correct columns on the DQB	Direct learners to the DQB (a large chart or wall display maintained across all 8 lessons). Say: 'In Lesson 1 we put up questions about why distance alone is not enough. Today we have new evidence. Write one thing we NOW KNOW — it must involve a drawing or a notation — and one thing we STILL WONDER about vectors. You have 2 minutes to write your two sticky notes.' (WAIT TIME: 2 minutes individual writing.) Collect and post notes as learners bring them up. Read 3 'We now know' notes aloud and ask: 'Does everyone agree with this claim?	Public Knowledge-Building Board (DQB Protocol): The DQB makes the class's collective growing understanding visible and persistent across lessons. By requiring that 'We now know' notes contain a specific geometric or notational claim, the teacher ensures learners are anchoring new abstractions (notation, equivalent vectors) to concrete evidence from the Observe and Explain phases. The 'We still wonder' notes generate authentic inquiry momentum toward Lesson 3 (vector addition), maintaining the storyline's coherence.	Teacher reads all posted sticky notes after class. Indicators of strong understanding: 'We now know' notes that correctly use vector notation and reference either the Nairobi grid or the matatu phenomenon. Indicators of incomplete understanding: notes that state 'a vector has direction' without linking to notation, or that confuse equivalent vectors with equal scalars. Use this information to plan the opening of Lesson 3 — address the most common misconception in the Lesson 3 Predict phase.

for similar readings	and briefly read two of their classmates' 'wonder' notes aloud.	Give me a thumbs-up.' For any 'wonder' note that anticipates Lesson 3 content (e.g., vector addition), say: 'That is a brilliant question — we are going to investigate exactly that in an upcoming lesson. Watch this board.' Point to the driving question at the top of the DQB: 'What mathematical tool do we need to describe a journey completely?' Ask: 'Are we closer to answering this today? How do you know?' Cold-call 2 learners. (WAIT TIME: 12 seconds each.)		
Model Building Phase 📹 **VIDEO:** [Reduction of alkynes] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 1 sketch-model (a rough diagram showing the two matatu routes with labels like 'distance' and 'direction'). They now upgrade this model using the tools from today: they replace any plain line segments with proper directed arrows, add vector notation labels (at minimum: column vector form and one symbolic form), mark the TAIL (Kencom) and HEAD (destination) of each journey-vector, and write a one-sentence model caption: 'The Kencom→Westlands journey is vector ____ and the Kencom→South B journey is vector ____, which are not equivalent because ____.' Learners compare their upgraded model with a partner and suggest one further improvement to each other's work.	Say: 'At the end of Lesson 1 you drew a sketch-model showing why the two matatu journeys are different. That model used words like direction and distance. Today you have better tools. Upgrade your model now — you have 4 minutes.' (WAIT TIME: 4 minutes independent model revision.) Circulate and use targeted prompts for learners who are stuck: 'Where is the TAIL of this vector? Where is the HEAD? What does the column vector for this journey look like?' After 4 minutes, say: 'Share with your partner. Tell them one thing their upgraded model shows clearly and one thing that could be even more precise.' Allow 90 seconds of partner feedback. Close by displaying a 'teacher model' on the board — a clean Nairobi grid with both matatu-journey vectors drawn, labelled in column form and symbolic notation, tails at Kencom, heads at Westlands and South B respectively. Say: 'This is what our model looks like after Lesson 2. In coming lessons, we will add more tools — and our model will grow. That is how mathematics works:	Iterative Model Revision (NGSS Practice 2 — Developing and Using Models): Requiring learners to physically upgrade an existing model — rather than draw a new one — makes the progression of understanding visible and felt. Comparing the Lesson 1 sketch (vague labels) with the Lesson 2 revised model (precise arrows and notation) gives learners tangible evidence of their own growing competence, building self-efficacy (a KICD core competency for this sub-strand). The partner-feedback step embeds peer formative assessment within the model-building process.	Teacher collects or photographs 4–5 upgraded models (spread across ability levels). Check for: (1) Correct arrowhead at the destination (HEAD), tail at Kencom. (2) Column vector notation correctly written with parentheses and correct component signs (negative for west/south components). (3) Model caption that correctly states the two vectors are NOT equivalent and gives a reason invoking both magnitude and direction (or direction alone if magnitudes are equal). Models that only show arrows without notation indicate the learner has grasped geometric representation but not yet symbolic representation — flag for targeted support at the start of Lesson 3.

		we build understanding piece by piece, just like a matatu driver learning every route in Nairobi.'		

D. TEACHER REFLECTION

1. During the Observe phase, did all learners successfully place arrowheads at the HEAD (destination) rather than the tail? If arrowhead-direction errors were common, how will I address this explicitly at the start of Lesson 3 before introducing vector addition?
2. Could learners articulate — in their own words — why two vectors in different positions on the Cartesian grid can be equivalent? Did the cut-out activity generate the 'same magnitude, same direction' language, or did learners rely on visual appearance? What follow-up is needed?
3. Were learners able to move between at least two notation forms (e.g., column vector and bold/underline) accurately? Which notation caused the most confusion, and how can I provide an additional reference chart or mnemonic for Lesson 3?
4. Looking at the DQB 'We still wonder' notes: what questions did learners generate, and do they align with the planned Lesson 3 content (vector addition)? Are there unexpected lines of inquiry that I should incorporate or address?
5. When learners upgraded their models in the Model Building phase, did the model caption task reveal any persistent scalar thinking (e.g., comparing the two journeys only by length)? How will I use those specific misconceptions as a launch point in the next lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that the two matatu journeys from Kencom — both 8 km long — produce two visibly distinct arrows on the Nairobi grid: one pointing north-west toward Westlands and one pointing south toward South B. They observed that moving a vector arrow to a new position on the Cartesian grid does not change the vector itself, as long as direction and length are preserved.
What did I learn?	A vector is represented geometrically as a directed line segment (arrow) with a TAIL (starting point) and HEAD (arrowhead at the destination). Vectors can be written in bold (v), underline ($\underline{v}$), column/component form, or AB-arrow notation. Two vectors are equivalent if and only if they have the same magnitude AND the same direction — position on the grid is irrelevant.
How does this explain the phenomenon?	The phenomenon is now mathematically visible: the Kencom→Westlands journey-vector and the Kencom→South B journey-vector are drawn as two distinct arrows on the grid — different directions, different endpoints — even though both have magnitude 8 km. This explains concretely why the matatu drivers end up at different destinations: their journey vectors are not equivalent. The number '8 km' alone (a scalar) cannot distinguish them, but the directed arrow — a vector — can.

LESSON 3 (80 minutes): Adding Vectors: Finding the Resultant of a Two-Leg Matatu Journey	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate how two successive displacement vectors — each with magnitude AND direction — combine to produce a single resultant vector, directly explaining why the matatu's final destination depends on more than just distance travelled.
Knowledge	Learners will know: (a) the triangle law of vector addition and the parallelogram law of vector addition; (b) how to add vectors geometrically on a grid using tip-to-tail placement; (c) how to add column vectors algebraically by adding corresponding components; (d) that the resultant vector is the single displacement vector equivalent to two or more successive displacements.
Skills	Learners will be able to: (a) represent two successive displacement vectors on a grid using directed line segments; (b) construct the resultant vector using the triangle law (tip-to-tail) and the parallelogram law; (c) add column vectors algebraically; (d) interpret the resultant vector as the single journey from start to final destination.
Attitudes	Learners will appreciate that direction is as mathematically essential as magnitude when describing real-world journeys, and will value collaborative sense-making as they build geometric and algebraic understanding of vector addition.
Key Inquiry Question	1. What is the meaning of the terms vector and scalar quantities? 2. Where are vectors used in real-life situations?
Purpose in Storyline	In Lessons 1–2 learners established that distance alone (a scalar) cannot explain why two 8 km matatu journeys from Kencom end at Westlands and South B respectively — they need magnitude AND direction, i.e. a vector. In Lesson 3, learners now apply that insight: they combine two sequential displacement vectors for the route Kencom → University of Nairobi → GPO using the triangle law and parallelogram law on grids, and add column vectors algebraically, to find the RESULTANT — the single vector that takes a passenger directly from Kencom to GPO. This resultant concept directly answers the driving question: the matatu's final destination is completely determined by the resultant of all displacement vectors along its route, not merely by total distance.
Safety Notes	No significant physical safety concerns. Ensure compasses/rulers used for geometric constructions are handled carefully. If learners move desks for group activity, remind them to lift rather than drag to reduce noise and injury risk.

B. LESSON OVERVIEW

Lesson 3 is the EXPLAIN + ADD VECTORS lesson in the 8-lesson Vectors sequence for Grade 10 Core Mathematics. Building on the prior two lessons in which learners defined vectors, distinguished them from scalars, and represented single vectors on grids and Cartesian planes, this lesson advances the storyline by asking: what happens when a matatu makes more than one leg of a journey? Learners work with a simplified Nairobi grid map showing the two-leg route Kencom Bus Stop → University of Nairobi → GPO. They first predict the resultant displacement by inspection, then construct it geometrically using the triangle law (tip-to-tail) and the parallelogram law on squared paper, and finally add the corresponding column vectors algebraically. The lesson closes by updating the Driving Question Board and revising the class vector model, firmly connecting the resultant vector to the anchoring phenomenon: the reason one matatu ends at Westlands and another at South B is precisely that their resultant displacement vectors — not just their distances — point to different final positions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a simplified Nairobi street grid (printed or projected) with two legs of a matatu journey marked as directed arrows: Leg 1 from Kencom Bus Stop to the University of Nairobi gate (labelled vector **a**) and Leg 2 from the University of Nairobi gate to GPO (labelled vector **b**). Both vectors are drawn on a square grid so their components are readable. Learners are individually asked to sketch, without any calculation, the single arrow they think would take a passenger DIRECTLY from Kencom to GPO in one straight trip — their predicted resultant. They annotate their sketch with a written sentence: 'I think the direct journey from Kencom to GPO goes ___ because ___.' Learners then share predictions with a partner (Think-Pair-Share, 3 minutes).	Display the Nairobi grid map prominently. Say verbatim: 'We have been saying a vector needs magnitude AND direction. Here is a matatu that leaves Kencom, goes to the University of Nairobi, then turns and goes to GPO. Without any maths yet — just your instinct and what you know about Nairobi — draw the ONE arrow that would take someone straight from Kencom to GPO. You have 3 minutes, working alone.' Issue squared paper. After 3 minutes, cold-call 3 learners to describe their predicted arrow (direction and rough length) — choose one who predicted a short arrow, one who predicted a long arrow, and one who predicted an unexpected direction. Record all three predictions visibly on the board under the heading 'OUR PREDICTIONS'. Ask: 'What information did you use to make that prediction?' Wait 12 seconds before taking responses. Do NOT confirm or correct predictions yet — explicitly say: 'We are going to test all three of these ideas today.'	Think-Pair-Share Prediction: By committing to a sketch before instruction, learners activate and expose their intuitive understanding of displacement. Displaying conflicting predictions creates productive cognitive dissonance — the class now has a concrete question to resolve through the geometric construction, directly motivating the triangle and parallelogram laws.	Circulate during individual sketching. Observable indicator: learner draws a directed arrow (not just a line) from Kencom to GPO. Listen for reasoning that references direction, not only distance. Flag any learner who draws a resultant that simply continues in the direction of Leg 2 (common misconception) — note these learners for targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES	Learners work in pairs with a printed A4 squared-paper grid on which vector **a** (Kencom → University of Nairobi) and vector **b** (University of Nairobi → GPO) are already plotted as directed line segments. The grid has a scale: 1 square = 100 m. Activity has two parts completed sequentially. PART A — Triangle Law (tip-to-tail): Learners are instructed to (i) confirm that vector **b** already starts at the tip of vector **a** (tip-to-tail arrangement), (ii) draw the resultant vector **r = a + b** as the arrow from the tail of **a** to the	Distribute grids and say: 'You are now going to find the ACTUAL resultant — not just predict it. Follow the steps on your activity card.' Circulate continuously. After 5 minutes, pause the class and address the most common error you observe by saying: 'I notice some groups are drawing the resultant in the wrong direction — remember, the resultant arrow goes from where the FIRST vector STARTS to where the LAST vector ENDS.' After Part A is complete (approximately 12 minutes), cold-call one pair to read their measured length and direction	Dual Representation Bridging: Learners construct the same resultant TWICE — once geometrically (tip-to-tail on grid) and once algebraically (column vector addition). Comparing the two representations develops the understanding that the resultant is a mathematical object with both a geometric meaning (directed arrow on a map) and an algebraic structure (sum of components). This dual encoding is essential for later work on position vectors and translation vectors.	Check: (1) Does the learner's drawn resultant arrow run from the tail of **a** to the tip of **b**? (2) Does the column vector addition show correct component-wise addition (not magnitudes added as scalars)? (3) Do the geometric and algebraic resultants agree? A learner who adds magnitudes only (e.g., writes |**r**| = |**a**| + |**b**|) is applying scalar thinking — intervene immediately by asking: 'If I walk 5 km north then 5 km south, what is my displacement? What is my distance? Are they the same?'

for similar readings	tip of **b**, (iii) use a ruler to measure its length in grid squares and a protractor to measure its direction from the horizontal. PART B — Column Vector Addition: Learners read off the column vectors for **a** and **b** from the grid (counting squares right/up for positive, left/down for negative), then add them algebraically: if **a** = (3, 4) and **b** = (5, −2) then **a + b** = (8, 2). They verify that the column vector (8, 2) matches the drawn resultant arrow on the grid. A worked example of the parallelogram law is shown on a class poster/slide for reference, where both vectors are redrawn from the same starting point (tail-to-tail) and the resultant is the diagonal of the completed parallelogram.	aloud. Write the values on the board. Then say: 'Now let us check that with column vectors' and lead Part B. Ask: 'What does the top number in a column vector tell us? What does the bottom number tell us?' Wait 10 seconds. Cold-call 2 learners. After Part B, ask: 'Does your algebraic answer (8, 2) match your drawn arrow? If not, where is the discrepancy?' Give 3 minutes for pairs to reconcile. Then show the parallelogram law poster and say: 'This is an alternative method — both laws give the SAME resultant. Why do you think that is?' Wait 12 seconds. Do not answer — note responses for Phase 3 sensemaking.		
Explain Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Whole-class sensemaking discussion anchored to the phenomenon. The teacher posts the original Nairobi map (from Lesson 1) showing the two 8 km routes — Kencom to Westlands and Kencom to South B — alongside the new grid showing the Kencom → UoN → GPO resultant. Learners are guided through three connected explanation tasks. TASK 1 — Generalise the triangle law: Learners write a definition of vector addition using the triangle law in their own words, then compare with a partner and agree on a class definition. TASK 2 — Apply to the phenomenon: Learners answer in writing: 'Using what you now know about vector addition and resultant vectors, explain in two sentences why the driver going to Westlands and the	Open the discussion by returning to the board predictions from Phase 1. Ask: 'Which prediction was closest to the actual resultant? Why?' Wait 12 seconds. Cold-call the learner whose prediction was closest and ask them to explain their reasoning. Then say: 'The key word we need today is RESULTANT. Who can tell me what the resultant vector represents in terms of the Kencom–GPO journey?' Wait 10 seconds. Cold-call 2 learners. Write the agreed definition on the board. For Task 2, say: 'Do not just say distance — use the words vector, direction, and resultant in your explanation.' After 5 minutes of writing, cold-call 3 learners to read their sentences. Facilitate peer critique: 'Does this explanation include direction? Does it use the word resultant	Anchored Explanation: By explicitly asking learners to use the resultant concept to re-explain the original phenomenon (two 8 km routes, different destinations), the teacher ensures that the new mathematical knowledge is not inert — it must do explanatory work on the problem that opened the unit. This is the EXPLAIN phase's core purpose: sensemaking is demonstrated when learners can apply the new construct to the anchoring context, not merely reproduce a procedure.	Task 2 written responses are the primary formative indicator. A proficient response explicitly states that: (a) both journeys have the same scalar distance (8 km) but (b) the displacement vectors point in different directions, so (c) the resultant displacement from Kencom is different in direction, placing the passenger at a different destination. An incomplete response that only references distance should prompt the teacher to ask: 'What direction does each matatu travel? Does that matter?' Collect Task 3 practice papers to review after the lesson.

	driver going to South B both travel 8 km but end at different places.' TASK 3 — Practise column vector addition: Three rapid independent practice problems using column vectors drawn from Kenyan city routes (e.g., a courier bike in Kisumu: leg 1 = (6, 3), leg 2 = (−2, 5); find the resultant). Learners solve independently then peer-mark.	correctly?' For the three practice problems, circulate and identify one learner who makes a sign error (negative component) — use this as a class teaching moment: 'What does a negative component mean on our Nairobi grid? Which direction is it?' Wait 10 seconds.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Area of a parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class Driving Question Board is displayed at the front. The DQB has the anchoring phenomenon description at the top and the driving question: 'Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?' Learners revisit sticky notes from Lessons 1 and 2 and do two actions: (1) UPDATE — move or annotate any prior sticky note that they can now answer more fully using today's language of 'resultant', 'triangle law', 'column vector addition'. (2) ADD NEW — add at least one new sticky note with either a new question that today's lesson has raised OR a new piece of evidence that helps answer the driving question. Examples of target new notes: 'The resultant tells you the straight-line journey from start to finish.' 'What if you add three vectors — can you still use triangle law?' 'Does the ORDER of addition matter — a + b vs b + a?'	Say: 'Take 4 minutes. Look at the DQB. What can you now answer that you could not answer at the start of this lesson? What new questions does today's activity raise?' Distribute sticky notes. After 4 minutes, cold-call 3 learners to share their new notes aloud and physically place them on the board. For each note, ask the class: 'Is this EVIDENCE that helps answer our driving question, or is it a NEW QUESTION we still need to investigate?' Sort notes into two columns on the DQB: 'What We Now Know' and 'What We Still Need to Find Out.' Highlight the note(s) about the resultant explaining final destination and say: 'This is the heart of today's lesson. By the end of our 8 lessons, every question on this board should have an answer.'	Driving Question Board Curation: The DQB functions as a visible, cumulative knowledge artefact. Physically moving notes from 'unknown' to 'known' gives learners a concrete representation of their growing understanding. The act of generating new questions (e.g., commutativity of vector addition, three-vector chains) previews future lessons and keeps the unit storyline coherent across all 8 lessons.	Read sticky notes for quality of mathematical language. A developing learner writes 'the arrow shows where you end up.' A proficient learner writes 'the resultant vector has the same starting point as the first vector and the same endpoint as the last vector, regardless of the path.' A learner who adds a question about commutativity (does **a + b = b + a**?) is demonstrating extended mathematical curiosity — note this for a stretch task in Lesson 4.
Model Building Phase 📹 **VIDEO:**	Learners return to the cumulative class vector model begun in Lessons 1–2. In Lesson 1 the model showed a single labelled	Say: 'Your model has been growing each lesson. Today you are adding the most powerful idea yet — the resultant. Open your	Cumulative Model Revision: The model-building phase operationalises the NGSS practice of Developing and Using Models.	Collect or photograph learner models at the end of the lesson. Look for three specific indicators: (1) The resultant arrow is drawn

Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	arrow (one vector) on a Nairobi grid. In Lesson 2 it was extended to show equivalent vectors and vector notation. Today, learners update their individual model diagram by: (1) Adding a two-leg journey (Kencom → UoN → GPO) with both component vectors labelled as column vectors, (2) Drawing and labelling the resultant vector using the triangle law, (3) Writing the column vector equation **r = a + b** alongside the diagram, (4) Adding a one-sentence caption: 'The resultant vector gives the direct displacement from the start to the final destination, regardless of the path taken.' Learners who finish early attempt to add a THIRD leg (GPO → Haile Selassie Avenue junction) and find the new resultant of all three legs.	model diagram from Lesson 2 and add today's Kencom–UoN–GPO example with the full column vector working shown.' Circulate and check that: (a) arrows are directed (not just lines), (b) the resultant arrow runs from the tail of the first vector to the tip of the last, (c) the column vector sum is written algebraically next to the diagram. After 8 minutes, hold a 3-minute gallery-walk: learners post models on desks and walk to view two peers' models. Ask them to leave a sticky dot on any model that shows the resultant correctly. Close by saying: 'Your model is now a tool that can EXPLAIN the phenomenon. Next lesson we will use this model to explore what happens when we multiply a vector by a scalar — and we will meet a new Nairobi route to test our ideas.'	Because the model grows across all 8 lessons, each addition is constrained by prior work — learners cannot simply copy; they must integrate new concepts (resultant, triangle law, column addition) into an existing representation. The gallery-walk peer review introduces a low-stakes accountability structure that surfaces errors (missing arrowheads, scalar addition of magnitudes) for immediate self-correction.	correctly (tail of **a** to tip of **b**). (2) The column vector equation is written in the form **r = a + b = (x_1+x_2, y_1+y_2)**. (3) The caption correctly uses the word 'displacement' (not 'distance'). Any model that labels the resultant with a scalar magnitude only (e.g., '8.25 km') without a direction indicates persistent scalar thinking — flag for follow-up in Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 2, did learners successfully reconcile their geometric (drawn) resultant with their algebraic (column vector) resultant? If several groups had discrepancies, was the source a counting error on the grid, a sign error in the column vector, or a conceptual misunderstanding of tip-to-tail placement? Plan a 5-minute recap at the start of Lesson 4 if more than one-third of the class showed discrepancies. (2) In Phase 3, Task 2 written explanations — did learners use the word 'direction' AND the word 'resultant' in their sentences? If not, the transition from scalar to vector thinking is incomplete and needs reinforcement through additional Kenyan context examples (e.g., a fisherman on Lake Victoria who paddles 3 km east then 4 km north — what is his resultant displacement from the starting point?). (3) Did the DQB generate new learner questions about commutativity (**a + b** vs **b + a**) or three-vector chains? If yes, these questions can be used as the entry point for Lesson 4's predict phase, maintaining learner intellectual ownership of the unit storyline. (4) Were all learner groups able to read column vectors from the grid without teacher support? If grid-reading was a barrier, prepare a short 'grid coordinates to column vector' reference card for struggling learners in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	On a squared-paper Nairobi grid, learners drew two successive displacement vectors — Kencom Bus Stop to University of Nairobi (vector a) and University of Nairobi to GPO (vector b) — and constructed the resultant vector using the triangle law (tip-to-tail), measuring its length and direction on the grid.
What did I learn?	The resultant vector is the single displacement vector from the tail of the first vector to the tip of the last vector; it can be found

	geometrically using the triangle law (tip-to-tail) or the parallelogram law, and algebraically by adding the corresponding components of column vectors (r = a + b).
How does this explain the phenomenon?	The reason two matatu drivers both travelling 8 km from Kencom CBD arrive at completely different destinations (Westlands vs South B) is that their displacement vectors point in different directions — the resultant of each driver's journey vectors, not the scalar total distance, determines the final destination.

LESSON 4 (80 minutes): Scaling the Journey: Scalar Multiplication and Subtraction of Vectors	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to multiply vectors by scalars (positive, negative, and fractional) and interpret the geometric and physical meaning of the resulting vectors, including vector subtraction as the addition of a negative vector.
Knowledge	Learners will know that: (i) multiplying a vector by a positive scalar changes its magnitude but preserves its direction and line of action; (ii) multiplying by a negative scalar reverses the direction; (iii) multiplying by a fraction scales the magnitude down; (iv) vector subtraction a − b is equivalent to a + (−b).
Skills	Learners will be able to: (i) draw scaled vectors on a grid given a scalar multiplier; (ii) determine the magnitude of a scalar multiple of a vector; (iii) express scalar multiples in column vector notation; (iv) subtract vectors geometrically and algebraically using the negative-vector concept.
Attitudes	Learners will appreciate that scalar multiplication is a powerful tool for modelling repeated or reversed journeys in real-life transport contexts such as boda-boda routes, and will show responsibility by working accurately with grid diagrams.
Key Inquiry Question	1. What is the meaning of the terms vector and scalar quantities? 2. Where are vectors used in real-life situations?
Purpose in Storyline	In Lessons 1–3 learners established that a journey requires both distance AND direction (the anchoring phenomenon), and practised representing and adding vectors. Lesson 4 now asks: what if a boda-boda rider makes the SAME trip 3 times, or turns around and rides back? Scalar multiplication formalises the idea of stretching, shrinking, or reversing a displacement vector — deepening the claim that a vector is a complete description of a journey. Subtraction of vectors is introduced as 'adding the reverse journey', directly connecting to the Nairobi map phenomenon where travelling −d from a destination returns you to the origin.
Safety Notes	No physical hazards in this lesson. Ensure learners use rulers carefully when drawing on grids. Remind learners not to share sharp pencils without care.

B. LESSON OVERVIEW

Learners begin by predicting what happens to a boda-boda displacement vector when the rider makes the same trip 3 times, half the trip, or rides back to the start. They then observe drawn grid examples and complete guided constructions of kd for k = 3, k = −1, k = −2, and k = ½, using the vector d mapped on a Nairobi-style street grid. In the Explain phase, pairs connect their drawings to column-vector arithmetic and verify magnitude changes. The DQB is updated to capture the new claim: 'Scalar multiplication scales magnitude and can flip direction.' A cumulative model is revised to show that a complete journey description now includes the possibility of repeated or reversed trips. The lesson advances the driving question by showing that the same 8 km vector from Kencom can be tripled, halved, or reversed — and the destination changes accordingly.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown the Nairobi street-grid anchor map (introduced	1. Display the anchor map and point to vector d. Say: 'We have	Think-Pair-Predict: activates prior knowledge of scaling from	Observable indicator: learner sketches show an attempt to

📹 **VIDEO:** [Video: Proportion Playground](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Multiplying Matrices by a Scalar (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	in Lesson 1) with vector d drawn from Kencom Bus Stop to Westlands. The teacher poses the boda-boda scenario verbally and in writing on the board: 'Kamau the boda-boda rider travels vector d from Kencom to Westlands. What vector describes: (a) Kamau making the SAME trip 3 times in a row from the same start? (b) Kamau making HALF the trip? (c) Kamau turning around and riding back from Westlands to Kencom?' Learners individually sketch their predictions on a mini-grid in their exercise books, labelling each predicted vector with a name. They then turn to a partner (Think-Pair) and compare sketches for 90 seconds before the teacher cold-calls three pairs. No confirmation or correction is given yet — predictions are posted on the DQB under the column 'Our Predictions'.	been describing Kamau's single trip as vector d. Today I want us to think about what happens when that trip is repeated, shortened, or reversed.' 2. Read all three sub-questions aloud slowly. Give 10 seconds of WAIT TIME in silence before learners begin sketching. 3. Circulate for 4 minutes — observe whether learners attempt to extend the arrow length for (a), shorten it for (b), or draw a new arrow in the opposite direction for (c). Do NOT correct. Note 2–3 examples of different predictions to reference later. 4. After Think-Pair (90 seconds), cold-call exactly 3 pairs: 'Pair 1 — show me your sketch for (a) and tell us your thinking in one sentence.' Repeat for Pairs 2 and 3. 5. Pin/write the three most contrasting predictions on the DQB. Say: 'Hold your ideas — we are about to gather evidence.'	everyday contexts (e.g. tripling a recipe of ugali) and creates cognitive dissonance around the direction-reversal case. By externalising predictions on the DQB, learners have a public record to confirm or revise, making subsequent sensemaking more meaningful.	change arrow length for (a) and (b), and a reversed arrow for (c). Teacher notes which learners draw (a) as three separate arrows (incorrect concatenation) versus one scaled arrow — this reveals whether they understand scalar multiplication vs. vector addition. This information guides cold-call selection.
Observe Phase 📹 **VIDEO:** [Video: Proportion Playground](#) Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Multiplying Matrices by a Scalar (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed or copied Activity Sheet containing a Nairobi-style 1 cm grid with vector d already drawn as 2 units East and 1 unit North (column vector: $d = (2, 1)^T$). Task instructions guide them through four constructions in pairs: (A) Draw 3d — start from the same origin, count $3 \times 2 = 6$ units East and $3 \times 1 = 3$ units North. Label it. (B) Draw ½d — count 1 unit East and ½ unit North. Label it. (C) Draw −d — count 2 units West and 1 unit South. Label it. (D) Draw −2d — count 4 units West and 2 units South. Label it. After drawing, each pair completes an observation table: for each vector, record (i) the column vector	1. Distribute activity sheets. Say: 'You have 15 minutes to complete all four constructions and fill the table. Use a ruler. Each square on the grid represents 1 km on the Nairobi map.' 2. After 3 minutes, circulate and check that learners are measuring from the same origin for each vector — a common error. If seen, say to the group: 'Check — are you starting ALL vectors from Kencom? Scalar multiplication does not move the starting point.' 3. At the 8-minute mark, pause the class. Cold-call 1 learner to share their column vector for −2d. Ask the class: 'Thumbs up if you agree, thumbs sideways if unsure.'	Structured Observation Table: the four-column table (column vector, magnitude, direction, line of action) guides learners to notice patterns systematically rather than drawing conclusions from a single case. The Nairobi grid context keeps the abstract operation anchored to physical displacement, supporting transfer of the mathematical concept.	Observable indicators: (i) column vectors for all four cases are correctly written — $3d = (6,3)^T$, $\frac{1}{2}d = (1, 0.5)^T$, $-d = (-2,-1)^T$, $-2d = (-4,-2)^T$; (ii) magnitudes computed correctly using Pythagoras ($\|d\| = \sqrt{5} \approx 2.24$ km, $\|3d\| = 3\sqrt{5}$, $\|\frac{1}{2}d\| = \frac{1}{2}\sqrt{5}$, etc.); (iii) direction column correctly identifies −d and −2d as 'opposite to d'; (iv) line of action correctly marked 'yes' for all four. Teacher uses observation-table responses to identify learners who need support with the magnitude calculation before the Explain phase.

	form, (ii) the magnitude (using Pythagoras), (iii) same direction as d / opposite / neither, (iv) same line of action as d: yes/no. Learners then answer: 'For constructions A–D, which destination on the Nairobi grid would Kamau reach if he started at Kencom each time?'	Address misconceptions immediately. 4. At 12 minutes, prompt: 'Now look at your line-of-action column. What pattern do you notice?' Give 10 seconds WAIT TIME. Cold-call 2 learners. 5. At 15 minutes, ask pairs to write one sentence summarising what scalar multiplication does to a vector's direction. Circulate and read 3–4 sentences. Select one accurate and one partially accurate sentence to use in the Explain phase.		
Explain Phase 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class discussion to formalise findings from the Observe phase. Learners contribute the pattern statements they wrote, and the teacher scribes a class definition on the board: 'If k is a scalar and a is a vector, then ka is a vector with magnitude $\|k\|\times\|a\|$, same line of action as a, direction preserved if $k > 0$, and direction reversed if $k < 0$.' Learners then work individually on two application problems set in Kenyan contexts: (1) Akinyi the matatu tout records that her matatu covers vector $r = (3, -4)^T$ km from Kisumu Bus Park to the fish market at Dunga Beach. Write in column form: 2r, −r, ½r. Calculate the magnitude of each. (2) A maize delivery truck travels vector m from Eldoret grain store to a depot. If −m takes the truck back to Eldoret, and 3m overshoots, sketch all three on a grid and describe the destination in each case. Finally, the teacher introduces vector subtraction: 'a − b means a + (−b). To go from Westlands back to Kencom and then on to South B, we add the	1. Begin by displaying two sentences collected during Observe — one accurate, one partially accurate. Ask: 'Which of these better matches our evidence from the grid? Why?' Give 10 seconds WAIT TIME. Cold-call 3 learners by name. 2. Scribe the class definition collaboratively — ask learners to suggest each word/condition rather than simply stating it. Say: 'What should we say about the magnitude? … What about when k is negative? Give me the exact condition.' 3. For the Kisumu application problem, give 5 minutes of independent work. Circulate and note errors. Cold-call 1 learner to write 2r on the board. Ask a second learner: 'Check this — is the magnitude correct? Show us the Pythagoras calculation.' 4. Before introducing vector subtraction, pause and connect to the phenomenon: 'If the matatu from Kencom to Westlands is vector d, and the matatu from Kencom to South B is vector e, how would you describe the	Claim-Evidence-Reasoning (CER): learners state a claim ('scalar multiplication scales magnitude and flips direction for negatives'), cite evidence from their observation tables, and reason using the formal algebraic definition. This strategy moves learners from empirical observation to generalised mathematical understanding. The subtraction-as-negative-addition framing directly connects to the phenomenon, reinforcing that vectors describe complete journeys.	Observable indicators: (i) learners correctly write $2r = (6,-8)^T$, $-r = (-3,4)^T$, $\frac{1}{2}r = (1.5,-2)^T$ for Akinyi's problem; (ii) magnitudes computed as 2×5=10, 5, 2.5 (since $\|r\|=5$ by Pythagoras on $(3,-4)^T$); (iii) a − b correctly computed as $(2,-4)^T$ algebraically and verified geometrically. Teacher listens during partner verification for learners who confuse the order of subtraction (b − a vs a − b), flagging this as a point of emphasis in the DQB phase.

	reverse of the first journey to the second.' Learners practise: given a = $(5, 2)^T$ and b = $(3, 6)^T$, find a − b geometrically and algebraically.	journey FROM Westlands TO South B?' Give 15 seconds WAIT TIME. Take 2 responses. Then say: 'That journey is e − d, which equals e + (−d). We reverse d and add.' 5. For the algebraic subtraction practice, cold-call 1 learner to write the answer on the board. Ask the class: 'Is this the same result we get geometrically on the grid? Verify with your partner in 60 seconds.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class DQB (a large chart or section of the board maintained since Lesson 1). In small groups of three, learners are given a sticky note or a strip of paper and asked to: (i) revise or confirm the prediction they posted in Phase 1 — was the prediction correct? (ii) write one new CLAIM their group can now make about vectors, supported by today's evidence; (iii) write one new QUESTION today's lesson has raised for them. Groups post their claims and questions. The teacher facilitates a 3-minute gallery walk (learners move to read two other groups' postings) and then reconvenes to read 2 claims and 2 questions aloud. The teacher explicitly connects the updated DQB to the driving question: 'We started asking why two matatus travel 8 km and end up in different places. Today we added another layer — if we scale or reverse that 8 km vector, the destination changes in a precise, predictable way. That is the power of the vector.'	1. Point to the original DQB predictions from Phase 1. Say: 'Look at what you predicted at the start of the lesson. In your group, decide: were you right, partially right, or did you need to revise? Write your revised claim on the strip.' Give 3 minutes for group writing. 2. During the gallery walk, say: 'You have 2 minutes. Read at least two other groups' claims. Be ready to tell us one claim you found interesting and WHY.' 3. After reconvening, cold-call 2 learners: 'Read one claim from another group that surprised you or matched yours.' 4. Connect to the driving question explicitly: write on the board 'Today's addition to our answer: kd tells us exactly where Kamau ends up after a scaled or reversed trip.' 5. Flag any unresolved questions for Lesson 5: 'One group asked about adding vectors that do not lie on the same line — that is exactly where we are headed.'	Driving Question Board Curation: learners publicly revise their thinking, creating a visible record of growing understanding across the unit. The gallery walk promotes peer-to-peer learning and models scientific practice of evaluating evidence and claims. Connecting new claims to the anchoring phenomenon ensures coherence across the 8-lesson sequence.	Observable indicators: (i) group claims accurately state that scalar multiplication changes magnitude by factor $\|k\|$ and reverses direction for $k < 0$; (ii) revised predictions correctly reject the idea that 3d is three separate arrows; (iii) new questions are generative (e.g., 'What if the scalar is 0?' or 'Can we multiply two vectors together?'). Teacher collects all strips to review after class and identify conceptual gaps to address in Lesson 5.
Model Building	Learners individually update the cumulative vector model they have been building since Lesson 1. The	1. Say: 'Open your model from last lesson. Today you are adding two new branches. You have 10	Cumulative Model Revision: the model-building strategy makes the growth of understanding visible	Observable indicators: (i) model correctly shows scalar multiplication node with three

Phase				
📹 **VIDEO:** [Video: Proportion Playground] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Multiplying Matrices by a Scalar (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	model is a diagram/concept map in their exercise book showing everything they now know about vectors. Today's additions required: (i) add a new node 'Scalar Multiplication' connected to the existing 'Vector' node, with the rule: kd has magnitude $\|k\|\|d\|$, direction same ($k>0$) or opposite ($k<0$), same line of action; (ii) add 'Vector Subtraction: $a - b = a + (-b)$' as a connected operation, linked to the 'Addition of Vectors' node from Lesson 3; (iii) annotate the Nairobi map sketch in the model — label the Kencom-to-Westlands vector as d, and add 3d (3 trips), −d (return), ½d (half trip) with arrows to approximate Nairobi destinations. Learners who finish early extend the model by adding: 'What is 0d? Draw it and describe it.' The lesson closes with the teacher asking one exit-ticket question verbally: 'If a vector $v = (4, -3)^T$, write down $-2v$ and state its magnitude without a calculator.'	minutes.' Display the three required additions on the board as a checklist. 2. Circulate for 8 minutes. For any learner who has only drawn the scalar multiplication rule but not annotated the Nairobi map, prompt: 'Can you show me where 3d ends up on your Nairobi sketch? Which direction is that from Kencom?' 3. At 10 minutes, cold-call 1 learner to describe their model update in two sentences to the class. Ask: 'Did anyone add something different or extra? Tell us.' 4. Pose the exit ticket verbally: 'Last question — no writing yet. Think silently for 10 seconds.' After the wait time: 'Now write your answer in your book. You have 60 seconds.' Cold-call 1 learner to share. Confirm: $-2v = (-8, 6)^T$, magnitude = $\sqrt{(64+36)} = \sqrt{100} = 10$. 5. Close with the lesson arc: 'From Lesson 1 we learned that distance alone is not enough. Today we learned that when we scale a vector, the direction is preserved or flipped — and the destination changes in a mathematically precise way. That is exactly the tool we are building.'	and personally owned. Requiring learners to connect new nodes (scalar multiplication, subtraction) to existing ones (vector definition, addition) reinforces the coherence of the vector concept rather than treating each lesson as isolated content. The Nairobi map annotation maintains the phenomenon connection throughout the model.	cases ($k>0$, $k<0$, $0<k<1$); (ii) subtraction correctly linked to addition-of-negative; (iii) Nairobi map annotation shows at least two correctly placed scaled vectors; (iv) exit ticket: $-2v = (-8,6)^T$ and magnitude = 10 stated correctly. Any learner who writes magnitude = −10 or confuses −2v with 2v reveals a residual misconception about the effect of negative scalars on direction vs. magnitude, to be addressed at the start of Lesson 5.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners successfully distinguish between multiplying by a negative scalar (reverses direction, same line of action) and adding a negative vector ($a - b$)? Were these two ideas conflated? (2) How well did learners connect the grid constructions back to the Nairobi matatu phenomenon — could they articulate WHERE 3d or −d would take Kamau on the Nairobi map? (3) Review the exit tickets: how many learners correctly computed the magnitude of −2v as 10 (not ±10)? If more than 30% made errors, plan a 5-minute review at the start of Lesson 5. (4) Check the DQB strips: are learners' new questions productive for Lesson 5 (position vectors, column vectors in 2D)? Adjust the Lesson 5 opening accordingly. (5) Note any learners who extended the model to ask about 0d — these learners are ready for the concept of a zero vector, which can be introduced informally in Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	On a Nairobi-style 1 cm grid, learners drew four scaled versions of displacement vector $d = (2,1)^T$: the vectors $3d$, $\frac{1}{2}d$, $-d$, and $-2d$, starting from Kencom Bus Stop. They recorded each result in a table showing the column vector, magnitude, direction relative to d, and line of action.
What did I learn?	Multiplying a vector by a scalar k changes its magnitude by the factor $\|k\|$ but keeps it on the same line of action. A positive k preserves the direction; a negative k reverses it. Vector subtraction $a - b$ is the same as adding the negative vector: $a + (-b)$. These rules allow us to describe repeated, halved, or reversed boda-boda and matatu journeys precisely.
How does this explain the phenomenon?	The results are explained by the definition of scalar multiplication: kd is a vector whose components are each multiplied by k, so its magnitude scales by $\|k\|$ and its direction is determined by the sign of k. Subtraction works because $-b$ is a vector of the same magnitude as b but in the opposite direction, and adding it to a gives the vector from the tip of b to the tip of a.

LESSON 5 (80 minutes): Column Vectors & Position Vectors — Giving Every Nairobi Landmark an Address	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to express vectors in column form on a Cartesian plane and to determine position vectors relative to the origin, linking geometric arrow representations to algebraic notation.
Knowledge	Learners will know that a column vector expresses a displacement as a vertical array of two components (horizontal and vertical), that a position vector describes the location of a point relative to a fixed origin O, and that the vector from point A to point B can be derived as $\vec{AB} = \vec{OB} - \vec{OA}$.
Skills	Learners will be able to: (g) determine column vectors in different situations; (h) determine position vectors in different situations; and apply these to plot Nairobi landmarks on a coordinate grid and derive displacement vectors between them.
Attitudes	Learners will appreciate that algebraic notation (column vectors) is a powerful and precise tool for describing real journeys — such as matatu routes from Kencom — and that direction encoded in road infrastructure is mathematically capturable.
Key Inquiry Question	1. What is the meaning of the terms vector and scalar quantities? 2. Where are vectors used in real-life situations?
Purpose in Storyline	In Lessons 1–4 learners established why direction matters (the Kencom phenomenon), explored geometric representation of vectors, equivalent vectors, vector addition, and scalar multiplication. In Lesson 5 the EXPLAIN phase formalises the algebraic language: learners translate the geometric arrows on the Nairobi grid into column vectors, assign position vectors to landmarks (Kencom = O, KICC = A, Uhuru Park = B), and derive $\vec{AB} = \vec{OB} - \vec{OA}$, directly answering why the two matatu drivers ended at different places — their displacement column vectors are completely different even though their magnitudes are the same.
Safety Notes	No physical hazards. Ensure graph paper/squared exercise books are available for every learner. If using printed Nairobi coordinate grid handouts, handle paper carefully to avoid cuts. Learners working in pairs should share rulers without conflict.

B. LESSON OVERVIEW

Lesson 5 is the primary EXPLAIN lesson in the 8-lesson vectors sequence. Learners move from geometric arrow representations (Lessons 1–4) to algebraic column vector notation. Using a simplified Cartesian grid of central Nairobi — where O = Kencom Bus Stop (origin), A = Kenyatta International Conference Centre (KICC), and B = Uhuru Park — learners plot position vectors $\vec{OA}$ and $\vec{OB}$, write them as column vectors, and derive the displacement vector $\vec{AB} = \vec{OB} - \vec{OA}$. The lesson resolves the core puzzle of the anchoring phenomenon: the two matatu drivers' journeys can now be written as distinct column vectors with the same magnitude but different components, proving that distance (scalar) alone cannot distinguish them — only the full column vector can. Learners update the Driving Question Board and revise their cumulative vector model of Nairobi. Core competencies developed include Learning to Learn and Self-Efficacy, with values of Responsibility and Unity embedded throughout.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown the Nairobi coordinate grid (Kencom at origin).	Display the Nairobi coordinate grid on the board (or distribute printed	Think-Pair-Share Prediction: activates prior knowledge of	Observable indicator: Learner written predictions include

📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Two-Column Proofs (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Two points are marked: A (KICC) and B (Uhuru Park). Learners are asked — WITHOUT instruction — to write in their exercise books: (a) How would you give someone precise written directions from Kencom to KICC using only numbers? (b) If a matatu travels 3 blocks east and 4 blocks north, and another travels 5 blocks east and 0 blocks north, are these journeys the same? Why or why not? Learners work individually for 4 minutes then share with a partner for 2 minutes (Think-Pair).	copies). Say: 'We have been drawing arrows to show vectors. Today I want you to think — how would you write a vector using only numbers, without drawing anything at all? Look at the grid. Kencom is at the origin. KICC is marked at point A. Uhuru Park is marked at point B. Individually, write your prediction in your exercise book. You have 4 minutes — begin now.' [WAIT — circulate silently for 4 minutes, note 3–4 interesting responses without correcting.] After 4 minutes: 'Share your idea with your neighbour. You have 2 minutes.' [WAIT 2 minutes.] Cold-call 3 learners: 'Wanjiku, tell us one idea your pair had.' / 'Otieno, did your pair agree or disagree — explain.' / 'Fatuma, what numbers did you think of using?' Do NOT confirm or deny — say: 'Interesting. Hold that idea. We are going to test it today.'	coordinates and direction from earlier lessons and creates cognitive dissonance — learners realise their informal language ('3 blocks east, 4 blocks north') is almost a column vector but lacks formal notation. This surfaces the need the lesson will meet.	reference to both a horizontal AND a vertical component (even if informally stated). Teacher notes which learners mention only one direction (likely still thinking scalarly) — these learners need targeted support in Phase 3.
Observe Phase 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Two-Column Proofs (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners receive (or copy from board) a worked demonstration: the teacher plots point A = KICC at coordinates (2, 3) on the Nairobi grid and draws vector $\vec{OA}$ as an arrow from the origin. The teacher then shows the column vector notation side-by-side with the arrow. Learners then independently plot B = Uhuru Park at (−1, 2), draw $\vec{OB}$, and write down what they observe about how the arrow relates to the two numbers in the column. Learners also revisit the Lesson 1 matatu map: the two 8 km routes from Kencom are now overlaid on the coordinate grid, and learners are asked to read off the approximate column vectors for each route's endpoint.	Say: 'Watch carefully. I am going to show you how we write a vector algebraically — using a column of two numbers.' On the board/grid, plot O at (0,0) and A at (2,3). Draw the arrow $\vec{OA}$. Write the column vector notation: $\vec{OA}$ = (2 above 3, written as a vertical column). Say: 'The top number tells us how far we move horizontally — positive means east, negative means west. The bottom number tells us how far we move vertically — positive means north, negative means south. So from Kencom to KICC we move 2 units east and 3 units north.' [WAIT 10 seconds.] 'Now you try. Plot point B — Uhuru Park — at (−1, 2) on your grid. Draw the arrow $\vec{OB}$ from the origin. Then write the column vector for $\vec{OB}$. You have 3 minutes.' [WAIT 3	Annotated Demonstration followed by Parallel Construction: the teacher models one example with explicit narration, then learners immediately replicate the process for a second point. Overlaying the original phenomenon map onto the coordinate grid directly connects new algebraic notation back to the anchoring puzzle — learners can now see WHY the two matatu routes are different: their column vectors are completely distinct despite both having magnitude 8 km.	Observable indicator: Learner's column vector for $\vec{OB}$ is correctly written as a vertical array with −1 on top and 2 below, with the arrow on the grid starting at O and ending at (−1, 2). Misconception to watch: learners who write the horizontal and vertical components in the wrong order (vertical first) — address immediately by pointing back to the 'east/west first, north/south second' convention and the road grid analogy.

		minutes — circulate, check arrow direction and column notation.] After 3 minutes, cold-call 2 learners to share their column vector on the board. Then display the original Lesson 1 matatu map overlaid on the coordinate grid. Say: 'Look at the two matatu routes we have been studying all week. Route 1 ends near coordinates (6, 5) — roughly Westlands. Route 2 ends near (−4, −6) — roughly South B. Write the column vectors for both endpoints in your books. What do you notice?' [WAIT 15 seconds.] Cold-call 2 learners.		
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Two-Column Proofs (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs to complete a structured task: (1) Given O = Kencom (0,0), A = KICC (2,3), B = Uhuru Park (−1,2), write $\vec{OA}$ and $\vec{OB}$ as column vectors. (2) Calculate $\vec{AB} = \vec{OB} - \vec{OA}$ by subtracting the components. (3) Verify by drawing the arrow directly from A to B on the grid and counting the grid squares — does it match the algebraic result? (4) Extension: A third landmark — C = Nairobi Railway Station at (−3, −2) — is introduced. Learners find $\vec{OC}$ and then $\vec{BC} = \vec{OC} - \vec{OB}$. (5) Whole-class share: pairs present their derivation of $\vec{AB}$ on the board and explain why the result makes sense in terms of the matatu journey.	Distribute or display the structured task. Say: 'Work in pairs. Use the three Nairobi landmarks on your grid. Your job is to find the column vector that takes you directly from KICC to Uhuru Park — we call this vector $\vec{AB}$. But instead of drawing and counting, I want you to calculate it algebraically using the position vectors. You have 8 minutes.' [WAIT — circulate, listen for pair discussions, prompt with: 'What does $\vec{OA}$ look like as a column? What does $\vec{OB}$ look like? What operation links them?'] After 8 minutes, cold-call one pair to write their working on the board. Say: 'Explain each step — what does each number mean in terms of the Nairobi map?' [WAIT 10 seconds for explanation.] Then ask the class: 'Does everyone's drawn arrow from KICC to Uhuru Park agree with the algebra? Thumbs up if yes, thumbs sideways if unsure.' Address any 'thumbs sideways' learners. Introduce the Railway Station extension: 'Now try point C — Nairobi Railway Station at (−3, −2). Find $\vec{BC}$. You	Worked-Example Pair Task with Geometric Verification: learners construct the algebraic result ($\vec{AB} = \vec{OB} - \vec{OA}$) and then verify it against the geometric grid drawing, building a two-way connection between algebraic and geometric representations. The deliberate return to the phenomenon (matatu drivers' column vectors) ensures the new knowledge is anchored in the driving question context, not treated as abstract computation.	Observable indicators: (1) Learner correctly writes $\vec{OA}$ = (2; 3) and $\vec{OB}$ = (−1; 2) as column arrays. (2) Learner correctly subtracts component-by-component: $\vec{AB}$ = (−1−2; 2−3) = (−3; −1). (3) Learner's drawn arrow from A(2,3) to B(−1,2) moves 3 units left and 1 unit down, matching the column vector. (4) Learner can articulate in words why the two matatu column vectors are different. Teacher watches for: sign errors in subtraction (common), and learners who subtract $\vec{OB} - \vec{OA}$ in the wrong order.

		have 5 minutes.' [WAIT.] Cold-call 2 different pairs for the extension. Consolidate by writing on the board: 'RULE: $\vec{AB} = \vec{OB} - \vec{OA}$ — the position vector of where you ARE minus the position vector of where you START.' Say: 'Now think back to our matatu drivers. Driver 1 ends at roughly (6, 5). Driver 2 ends at (−4, −6). Write their displacement column vectors from Kencom. Are they the same?' [WAIT 10 seconds.] Cold-call 1 learner: 'Kamau, what does this tell us about why they arrived at different places?' Expect and affirm: 'Even though the magnitudes might be similar, the column vectors are completely different — one goes east and north, the other goes west and south.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Two-Column Proofs (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	The class returns to the Driving Question Board displayed at the front of the room. Each pair receives two sticky notes. On the first (green): write one thing the column vector idea has now ANSWERED from the driving question — 'Why do two matatu journeys of exactly the same distance end up at completely different destinations?' On the second (orange): write one NEW question that arose today — e.g., 'Can we use column vectors to find the actual straight-line distance?' or 'What if there are more than two landmarks — can we chain position vectors?' Pairs post their notes on the DQB under the headings 'NOW WE KNOW' and 'WE STILL WONDER'. The teacher reads 3–4 notes aloud and makes explicit links to the lesson's learning.	Say: 'We started this unit with a puzzle — two matatu drivers, same distance from Kencom, completely different destinations. Take your green sticky note and write exactly HOW the column vector idea answers that puzzle. Be specific — use numbers from today's lesson.' [WAIT 3 minutes for pairs to write.] 'Now on your orange note, write one new question today's lesson has created for you.' [WAIT 2 minutes.] 'Come post your notes on the DQB.' [Allow 2 minutes for posting.] Read 2 green notes aloud: 'Listen to what Akinyi and Mwangi wrote: (read). Do we all agree this answers the driving question?' [WAIT 10 seconds — cold-call 1 learner for a response.] Read 2 orange notes: 'Interesting — Chebet is asking whether we can find the actual distance from a column vector. That is exactly	Driving Question Board Protocol with Colour-Coded Notes: the two-colour system forces learners to distinguish between resolved understanding (green) and generative new questions (orange), making metacognitive progress visible to both teacher and learners. Reading selected notes aloud validates learner thinking and creates a shared class narrative of growing understanding.	Observable indicator: Green notes contain specific reference to column vector components (not just 'direction matters') — e.g., 'Driver 1's vector is (6;5) and Driver 2's is (−4;−6) so they point in completely different directions even if the lengths are similar.' Orange notes that reference magnitude or distance indicate readiness for Lesson 6 content.

		what Lesson 6 will answer — the magnitude of a vector.' Point to the DQB and say: 'Notice how our board is filling up — every lesson we know more and wonder more. That is what real mathematical thinking looks like.'		
Model Building Phase 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Two-Column Proofs (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative 'Nairobi Vector Map' model started in Lesson 1 (a personal annotated map/grid built up across the sequence). Today they add: (1) A Cartesian coordinate grid overlay with O = Kencom as origin. (2) At least three position vectors ($\vec{OA}$ to KICC, $\vec{OB}$ to Uhuru Park, $\vec{OC}$ to Nairobi Railway Station) drawn as arrows AND written as column vectors beside each arrow. (3) One derived displacement vector (e.g., $\vec{AB}$) drawn and labelled with its column form and the subtraction working shown. (4) A written sentence at the bottom of their model: 'A column vector tells us both the magnitude AND direction of a displacement, which is why the two matatu drivers arrived at different places even though they both travelled 8 km.' Learners who finish early add $\vec{OD}$ to a fourth landmark of their choice (e.g., University of Nairobi at (1,−2)) and derive $\vec{DC}$.	Say: 'Get out your Nairobi Vector Map from Lesson 1. Today you are upgrading it — you are going to add the algebraic layer on top of the geometric layer.' Display a model example on the board showing what the finished upgrade should look like (coordinate grid, three position vector arrows, column vector notation, one derived displacement vector). Say: 'You have 10 minutes to complete the upgrade. Make it clear enough that someone who missed this lesson could pick it up and understand how to get from Kencom to any of these landmarks using column vectors.' [WAIT — circulate, prompt: 'Is your column vector written with the horizontal component on top?' / 'Have you shown the subtraction working for $\vec{AB}$?' / 'Can you explain your model to me in one sentence?'] At 8 minutes: 'Two minutes left — write your summary sentence at the bottom.' At 10 minutes, cold-call 2 learners to hold up their models and read their summary sentence aloud. Affirm and correct as needed. Close with: 'Next lesson we will use these column vectors to calculate the exact distance between any two landmarks — we will finally be able to give a matatu driver both a direction AND a distance in one mathematical object.'	Cumulative Model Revision: by physically updating the same map across all 8 lessons, learners build a tangible record of conceptual growth. The requirement to write both the geometric arrow AND the algebraic column vector side-by-side on the same model forces learners to connect both representations simultaneously, deepening understanding beyond rote notation practice.	Observable indicators: (1) Model shows correctly oriented arrows from origin to each landmark. (2) Column vectors are written in correct vertical array format with signs. (3) $\vec{AB}$ subtraction working is shown and result matches the drawn arrow. (4) Summary sentence correctly attributes the matatu phenomenon to the column vector components being different. Teacher collects 4–5 models at the end of class for written feedback before Lesson 6.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) PHENOMENON CONNECTION — Could every learner articulate, in their own words, why the two matatu drivers arrived at different destinations using column vector language? If not, which phase lost the connection and how will you re-anchor it in Lesson 6's opening? (2) NOTATION PRECISION — How many learners consistently wrote the column vector in the correct vertical array format (not horizontal, not comma-separated)? Plan a 3-minute notation drill for the start of Lesson 6 if more than a quarter of the class was inconsistent. (3) SUBTRACTION DIRECTION — Did learners confuse $\vec{OB} - \vec{OA}$ with $\vec{OA} - \vec{OB}$? This is the most common error in this sub-lesson. Consider creating a mnemonic: 'WHERE you are minus WHERE you started — just like matatu destination minus departure.' (4) DQB QUALITY — Review the orange 'wonder' sticky notes. Are learners' questions pointing naturally toward magnitude (Lesson 6) and translation (Lesson 7)? If most orange notes are about unrelated topics, the storyline thread may need reinforcing in the lesson opening. (5) INCLUSION — Were learners who were identified as 'scalar-only thinkers' in Phase 1 able to reach the column vector understanding by Phase 3? What additional scaffolding do they need?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	On a Nairobi coordinate grid with Kencom Bus Stop at the origin, we plotted position vectors $\vec{OA}$ to KICC at (2,3) and $\vec{OB}$ to Uhuru Park at (−1,2). We drew the arrows and placed the column vector notation beside each. We also looked at the two original matatu routes from Kencom — the Westlands route ending near (6,5) and the South B route ending near (−4,−6) — and wrote their column vectors side by side.
What did I learn?	A column vector writes a displacement as a vertical pair of numbers: the top number gives horizontal movement (positive = east, negative = west) and the bottom number gives vertical movement (positive = north, negative = south). A position vector $\vec{OA}$ gives the location of point A relative to origin O. To find the vector from A to B we use $\vec{AB} = \vec{OB} - \vec{OA}$, subtracting the components. The two matatu drivers' journeys have completely different column vectors — (6;5) versus (−4;−6) — which is exactly why they arrived at different destinations even if their distances were similar.
How does this explain the phenomenon?	Distance alone (a scalar) cannot distinguish two journeys that start at the same point — it tells you HOW FAR but not WHERE. A column vector captures BOTH the magnitude and direction of a displacement in a single algebraic object. Because the Westlands matatu moves predominantly east and north while the South B matatu moves west and south, their column vectors point in opposite directions. This is the mathematical reason two matatu drivers travelling the same distance from Kencom arrive at completely different destinations: their displacement column vectors are different, not just their distances.

LESSON 6 (40 minutes): Magnitude & Midpoint — How Far Is the Fishing Ground, and Where Is the Halfway Mark?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners derive the magnitude formula $\lvert v\rvert = \sqrt{x^2 + y^2}$ from Pythagoras' theorem and calculate the midpoint of a vector, connecting both ideas to Kenyan fishing and athletics contexts.
Knowledge	– The magnitude (length) of a vector v = (x; y) is given by $\lvert v\rvert = \sqrt{x^2 + y^2}$, derived directly from Pythagoras' theorem applied to the right-angled triangle formed by the horizontal and vertical components. – A scalar quantity (distance) can now be recovered from a vector quantity (displacement) — magnitude strips away direction and gives only 'how far'. – The midpoint M of a vector from point A to point B is the point $M = ((x_A + x_B)/2 ; (y_A + y_B)/2)$, representing the halfway position along the journey.
Skills	– Apply Pythagoras' theorem to calculate the magnitude of a given column vector in contextualised Kenyan problems. – Determine the midpoint of a vector given the coordinates of the start and end points, and interpret the result in a real-life context.
Attitudes	– Appreciate that a single mathematical formula (Pythagoras' theorem) learned earlier can be extended to unlock new meaning in a different mathematical context (vectors). – Value precision: recognise that knowing only the distance (scalar) is insufficient for navigation — direction and an exact halfway point matter to real fishermen and athletes.
Key Inquiry Question	Where are vectors used in real-life situations — and how do we measure how far a vector actually travels?
Purpose in Storyline	Lessons 1–5 established that vectors require both magnitude and direction, and gave learners tools to represent, add, subtract, scale, and write vectors in column form. Lesson 6 now asks: given a column vector, how long is it? And where is its midpoint? These two calculations let learners finally answer a lurking sub-question of the driving question — the '8 km' labels on the matatu map are the magnitudes of the two displacement vectors; by computing $\sqrt{x^2 + y^2}$ for each route's column vector, learners confirm (or discover) that equal magnitudes with different directions produce completely different destinations.
Safety Notes	No specific hazards. If learners use calculators, remind them to check square-root outputs by estimating mentally first to catch key-entry errors.

B. LESSON OVERVIEW

This lesson sits at the heart of the unit's evidence-gathering sequence. In previous lessons learners built up a rich toolkit: they can represent vectors as arrows, add and subtract them, multiply by scalars, write them in column form, and use position vectors to locate Nairobi landmarks. One question has been quietly growing: the matatu phenomenon introduced an '8 km' label — a scalar — yet the whole unit is about vectors. How does the number 8 km relate to the column vector that describes each route? Lesson 6 resolves this by showing that the magnitude of a vector is nothing more than the length of the hypotenuse of the right-angled triangle whose legs are the horizontal and vertical components — Pythagoras' theorem in a new disguise. Learners derive $\lvert v\rvert = \sqrt{x^2 + y^2}$ themselves through a structured guided-inquiry sequence anchored in a vivid Lake Victoria fishing scenario set off Kisumu pier.

The midpoint of a vector is introduced in the second half of the lesson through a Kenyan athletics track context: a relay runner sprints from point A to point B along a known displacement vector — where exactly should a coach stand at the halfway mark to give the baton-passing signal? Learners discover that midpoint coordinates are simply the mean of corresponding coordinates, a result that feels intuitive once they see it on a grid. Both formulas — magnitude and midpoint — are practised in paired problem-solving before learners return to the driving question map.

The lesson closes by inviting learners to recompute the magnitudes of the two matatu column vectors from Lesson 5 (the Westlands route ≈ (6; 5) and the South B route ≈ (−4; −6)) and verify that their magnitudes are indeed close to 8 km. This is a satisfying 'aha' moment that cements the distinction between the scalar distance and the vector displacement, and directly advances the driving question: same magnitude, completely different direction — that is why the two drivers ended up in different places.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Euclid as the father of geometry] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Resultant as Magnitude and Direction (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-page card showing a simple grid with a right-angled triangle drawn on it. The horizontal leg is labelled '6 km east' and the vertical leg '8 km north' — representing a fishing boat's journey from Kisumu pier. Learners are asked to write in their exercise books: 'Without using a ruler, predict the straight-line distance (the actual displacement distance) from the pier to the fishing ground. Explain your reasoning in one sentence.' After writing individually for 2 minutes, learners share predictions with a seat-partner and agree on a joint estimate.	Display the grid card on the board (drawn large with chalk or projected). Read aloud: 'A fishing boat leaves Kisumu pier, travels 6 km due east, then 8 km due north to reach a rich fishing ground on Lake Victoria. The skipper wants to know — if he could travel in a straight line from the pier to the fishing ground, how far would that straight-line distance be? Write your prediction now.' Allow 2 minutes of silent individual writing. Then say: 'Turn to your neighbour — share your prediction and your reason. You have 90 seconds.' After pair-share, cold-call 3 learners (choose one who looks confident, one uncertain, one who has drawn something in their book). For each response ask: 'What mathematical idea are you using to make that prediction?' Allow 10 seconds of wait time after each response before accepting or probing. Do NOT confirm or deny predictions — record them visibly on the board under the heading 'Our Predictions'.	Predict-then-Justify: Learners commit to a numerical estimate and articulate the reasoning behind it before instruction begins. This activates prior knowledge of Pythagoras' theorem from Junior School and creates productive cognitive tension when their estimate is later checked. Recording predictions publicly on the board holds learners accountable and makes the resolution moment more memorable.	Teacher listens for: (1) learners who immediately invoke Pythagoras — evidence of strong prior knowledge; (2) learners who add 6 + 8 = 14 — a common misconception (adding scalars rather than using the theorem); (3) learners who draw a right-angled triangle — evidence of geometric thinking. Note which misconception (linear addition) appears so it can be explicitly addressed in the Observe phase.
Observe Phase 📹 **VIDEO:** [Euclid as the	Learners open their exercise books to a fresh page and follow a structured guided derivation in four steps written on the board. Step 1:	Write the four derivation steps on the board one at a time, pausing between each. After Step 2 ask (cold-call, 2 learners): 'Why do we	Worked-Example Fading: The teacher fully works through the concrete fishing-boat example (numbers only), then fades the	Circulate during the 3-minute individual work period. Look for: (1) learners squaring correctly — $5^2 = 25$, $12^2 = 144$; (2) learners

father of geometry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resultant as Magnitude and Direction (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Draw the right-angled triangle for the fishing boat journey (horizontal leg = 6, vertical leg = 8). Step 2: Label the hypotenuse as $\|v\|$ — the straight-line displacement distance. Step 3: Apply Pythagoras: $\|v\|^2 = 6^2 + 8^2 = 36 + 64 = 100$, so $\|v\| = \sqrt{100} = 10$ km. Step 4: Now replace the numbers 6 and 8 with letters x and y and re-derive: $\|v\|^2 = x^2 + y^2$, therefore $\|v\| = \sqrt{x^2 + y^2}$. Learners then immediately apply the formula to a second scenario on a printed/chalked problem card: 'A fishing boat travels from Kisumu pier with displacement vector v = (−5; 12). What is the straight-line distance to the fishing ground?' Learners work individually for 3 minutes, then compare with a partner. A third problem introduces a non-integer answer: v = (3; 4) scaled — v = (9; 12) — so that learners see the formula still works and the answer simplifies to 15.	call this $\|v\|$ and not just v? What does the straight bar mean?' Wait 10 seconds each. After Step 3, point to the predictions board: 'Look at our predictions. Who predicted 10? Who predicted 14? Why does Pythagoras give us 10 and not 14?' Wait 12 seconds, then cold-call 1 learner who originally predicted 14 and ask them to explain the error in their own prediction — frame this positively: 'This is exactly the mistake many people make — can you explain to the class what went wrong?' After Step 4, say: 'Write the magnitude formula in a box in your notes — this is a formula you must be able to recall and use.' For the second fishing-boat problem with v = (−5; 12), after 3 minutes ask: 'Does it matter that the x-component is negative? Think for 10 seconds, then tell your partner.' Cold-call 1 pair: 'What happens when you square a negative number?' Confirm: squaring removes the negative sign, so $\|v\| = \sqrt{25 + 144} = \sqrt{169} = 13$ km. Write on board: '$\|v\| = \sqrt{x^2 + y^2}$ — direction disappears, only distance survives.'	numbers to algebra (x and y), so learners see the general formula emerge naturally from a specific case. This is followed immediately by two application problems where learners reproduce the process independently. The strategy builds procedural fluency while keeping conceptual understanding (why does the formula work?) central.	who write $(-5)^2 = -25$ — the key error to address; (3) learners who arrive at $\sqrt{169}$ but cannot evaluate it — they need a prompt: 'What number multiplied by itself gives 169?' Target feedback to learners making the sign error immediately rather than waiting for whole-class correction.
Explain Phase 📹 **VIDEO:** Euclid as the father of geometry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resultant as Magnitude and Direction (Flexbook)	The teacher introduces the midpoint concept through a vivid athletics scenario: 'Achieng is a relay runner at a national championships track meet in Nairobi's Kasarani Stadium. She starts at point A = (2; 3) and finishes her leg at point B = (10; 9) on a coordinate grid (units in metres). Her coach wants to stand exactly at the halfway point to shout the baton-passing cue. Where is that halfway point?' Learners draw the two points on a coordinate grid, then discuss in	Introduce the Kasarani scenario with energy: 'Achieng Atieno is running the 4×400 m relay at Kasarani. Her coach must stand at the exact halfway point of her leg to give the signal. Let us help the coach.' Draw the coordinate grid on the board with points A and B marked. Say: 'Before I give you any formula, discuss with your partner for 90 seconds: how would you find the halfway point just by looking at the grid?' After pair discussion, cold-call 2 pairs (wait 10 seconds each). Expect: count	Connect-Back Anchoring: After establishing the two new formulas (magnitude and midpoint) through fresh Kenyan contexts (Kisumu fishing, Kasarani athletics), learners are immediately redirected to the original driving question map to apply both formulas. This prevents isolated procedural learning and keeps every new piece of mathematics tethered to the phenomenon that motivated the entire unit. The explicit calculation showing $\|v_1\| \approx \|v_2\| \approx 8$ km is the 'reveal' moment	During the midpoint practice, look for: (1) learners who add coordinates without dividing by 2 — prompt: 'What does the word midpoint mean? How many equal parts?'; (2) learners who handle negative coordinates correctly in the second practice problem — this is a good indicator of solid understanding. During the matatu magnitude calculation, listen to partner discussions: can learners articulate in their own words why two vectors with similar magnitudes but different directions

Source: CK-12 🔍 Search ARES for similar readings	pairs how to locate the midpoint geometrically (count the grid squares to the middle). After 2 minutes the teacher guides the class to discover the midpoint formula: M = ((x_A + x_B)/2 ; (y_A + y_B)/2) = ((2+10)/2 ; (3+9)/2) = (6; 6). Learners then practise two further midpoint problems: (a) A = (0; 4), B = (8; 0) — midpoint of Achieng's warm-up sprint; (b) A = (−3; 5), B = (7; −1) — midpoint of a displacement vector from a market in Kisumu to a school. For the final 5 minutes of the Explain phase, learners return to the driving question map and compute both the magnitude AND the midpoint of the two matatu displacement vectors from Lesson 5: Westlands route v_1 = (6; 5) and South B route v_2 = (−4; −6), starting from the origin (Kencom Bus Stop).	half the horizontal distance, count half the vertical distance. Then say: 'Excellent. Let us write that as a formula.' Derive the midpoint formula step by step on the board. After the two practice problems, say: 'Now here is the big moment — take out your Lesson 5 notes. Find the column vectors for the Westlands route and the South B route. Calculate (a) the magnitude of each and (b) the midpoint of each journey.' Allow 4 minutes of paired work. Then cold-call 1 pair for magnitudes: '$\|v_1\| = \sqrt{6^2 + 5^2} = \sqrt{36 + 25} = \sqrt{61} \approx 7.8$ km — close to 8 km!' and '$\|v_2\| = \sqrt{(-4)^2 + (-6)^2} = \sqrt{16 + 36} = \sqrt{52} \approx 7.2$ km — also close to 8 km!' Write on board: 'Same (approximate) magnitude. COMPLETELY different direction. COMPLETELY different destination.' Pause 12 seconds to let this land, then ask the whole class: 'So what does the 8 km label on the matatu map actually tell us, and what does it NOT tell us?'	that makes the phenomenon make mathematical sense.	produce different destinations? This verbal articulation is the target understanding for the lesson.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Euclid as the father of geometry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resultant as Magnitude and Direction (Flexbook) Source: CK-12	Learners take 3 minutes to write on sticky notes (or small paper slips) responses to two prompts displayed on the board: (1) 'What NEW thing does today's lesson let us say about the matatu phenomenon that we could not say before?' and (2) 'What question does today's lesson raise for you that is still unanswered?' Learners stick their notes on the class DQB under two columns: 'We can now explain…' and 'We still wonder…'. The teacher reads out 3–4 responses from each column aloud, with no evaluation — just reading and acknowledgement.	Point to the DQB and say: 'We have been adding to this board since Lesson 1. Today is a big day — we can now actually calculate the 8 km. Take 3 minutes and write your two sticky notes.' Circulate and briefly read what learners are writing — do not correct at this stage. After 3 minutes, invite learners to come up and place their notes. Read 3–4 from the 'We can now explain' column: target notes that say things like 'we can prove both matatu routes are about 8 km using $\sqrt{x^2+y^2}$' or 'the matatu distances are the magnitudes of the column vectors'. Read 2–3 from the 'We still wonder' column:	Public Knowledge Construction on the DQB: The Driving Question Board functions as a living, communal record of the class's growing understanding. By separating 'what we can now explain' from 'what we still wonder', learners practise metacognitive monitoring — they must assess whether their current understanding is sufficient or whether gaps remain. The teacher's act of reading responses aloud without evaluation models intellectual safety and signals that wondering is as valued as knowing.	Teacher scans the 'We can now explain' notes for evidence that learners have connected magnitude to the phenomenon (not just stated the formula abstractly). A note that says only 'we learned $\|v\| = \sqrt{x^2+y^2}$' shows procedural recall; a note that says 'the 8 km on the matatu map is the magnitude of the displacement vector' shows conceptual connection. Count the ratio — if fewer than half the class have made the conceptual connection, plan to revisit the phenomenon connection at the start of Lesson 7.

🔍 Search ARES for similar readings		likely to include questions about what happens when vectors are in 3D, or how to find direction as an angle, or what translation vector is. Point to these 'still wonder' notes and say: 'These are excellent — they are showing us where we are going next in our unit. Hold on to these questions.'		
Model Building Phase 📹 **VIDEO:** Euclid as the father of geometry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Resultant as Magnitude and Direction (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open to their cumulative vector model — the annotated Nairobi street grid map they have been updating since Lesson 1. They now add two new layers: (1) They label the straight-line displacement arrow for each matatu route with its calculated magnitude ($\|v_1\| \approx 7.8$ km for the Westlands route, $\|v_2\| \approx 7.2$ km for the South B route), confirming these match the '8 km' labels in the original phenomenon. (2) They mark and label the midpoint of each route's displacement vector on the map, interpreting it physically: 'At this point on the map, the matatu is halfway through its straight-line displacement from Kencom.' Learners also write a two-sentence 'model caption' beneath their map: 'The 8 km labels are the magnitudes of the displacement vectors — they confirm how far each journey is in a straight line. The midpoint of each displacement vector shows the halfway position, which is different for each route because the directions are different.'	Say: 'Take out your matatu maps — the model we have been building together. Today you are going to add two new pieces of mathematical information to that model.' Give clear instructions: 'First, write the magnitude you calculated next to each route's displacement arrow — use the ≈ symbol because our column vectors were approximations. Second, find and mark the midpoint of each displacement arrow on your map, and label it with its coordinates.' Allow 4 minutes of individual work. Circulate and check: (1) Are learners placing midpoints geometrically in a plausible position (roughly halfway along each arrow)? (2) Are the magnitude labels consistent with the calculations done in the Explain phase? After 4 minutes, ask the class to hold up their maps. Scan for obvious errors and address 1–2 quickly. Then read the model caption prompt from the board and give 2 minutes to write it. Close the phase by saying: 'Look at the two midpoints on your map. Are they the same point? Why not? Discuss with your partner for 30 seconds.' Cold-call 1 pair: the expected answer is that the midpoints are different because the directions are different, even though the	Cumulative Model Revision: The matatu map functions as a running scientific/mathematical model that learners update with each lesson's new knowledge. Adding magnitude labels and midpoints to the map makes abstract calculations physically visible on a representation of real Nairobi geography. The model caption forces learners to synthesise — they must put into words what the new annotations mean for the phenomenon, not just what they calculated. This is an NGSS Science and Engineering Practice analogue: developing and using models.	Collect or photograph 4–5 learner maps at the end of the lesson. Check: (1) Are magnitude values correctly computed and placed on the correct route arrows? (2) Are midpoints correctly located — both coordinates correct? (3) Does the model caption accurately connect magnitude to the '8 km' phenomenon label? Use these maps to identify learners who need targeted support at the start of Lesson 7 before moving into translation vectors.

		magnitudes are similar. Affirm: 'This is the heart of our driving question — same magnitude, different direction, different everything — different destination, different midpoint, different journey.'		

D. TEACHER REFLECTION
1. When learners computed the magnitudes of the two matatu column vectors and found both were approximately 8 km, what was the quality of their 'aha' moment — did most learners make the connection to the phenomenon independently, or did it require explicit teacher prompting? What does this tell me about how well I am building the storyline across lessons? 2. Did the negative component in the second fishing-boat vector ($v = (-5; 12)$) cause widespread errors? If more than one-third of the class wrote $(-5)^2 = -25$, how should I address this algebraic prerequisite at the start of Lesson 7 without losing lesson momentum? 3. How effectively did the Kasarani athletics midpoint context resonate with learners — did it feel genuinely Kenyan and relatable, or did it feel forced? Would a different context (market stall layout, school compound) have worked better for this class? 4. Reviewing the DQB 'We still wonder' sticky notes: what are the most common questions learners are asking, and are they aligned with where Lesson 7 (translation vectors) is heading? Are there any misconceptions embedded in those questions that I need to address proactively? 5. Did learners' cumulative model maps show genuine integration of today's magnitude and midpoint labels with the column vectors and arrows from previous lessons, or are the maps becoming cluttered and hard for learners to read? Should I provide a cleaner base map for the final two lessons of the sequence?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that the right-angled triangle formed by the horizontal and vertical components of a column vector has a hypotenuse whose length is exactly the straight-line displacement distance — and that applying Pythagoras' theorem to the two matatu column vectors from Lesson 5 gives magnitudes of approximately 7.8 km and 7.2 km, confirming the '8 km' labels on the phenomenon map. Learners also observed, on a coordinate grid, that the midpoint of a displacement vector from A to B lies at the average of the corresponding coordinates.
What did I learn?	The magnitude of a column vector $v = (x; y)$ is $\lvert v\rvert = \sqrt{x^2 + y^2}$, derived from Pythagoras' theorem; this scalar value tells us how far a displacement is in a straight line but discards direction entirely. The midpoint M of the journey from point $A(x_A; y_A)$ to point $B(x_B; y_B)$ is $M = ((x_A + x_B)/2 ; (y_A + y_B)/2)$, giving the exact halfway position along the displacement. A negative component does not affect the magnitude because squaring always produces a positive value.
How does this explain the phenomenon?	The '8 km' labels on the matatu map are the magnitudes of the two displacement vectors — they confirm that both drivers travel the same straight-line distance from Kencom Bus Stop. However, magnitude alone cannot determine destination: the Westlands vector (6; 5) and the South B vector (−4; −6) have almost identical magnitudes (~7.8 km and ~7.2 km) but point in completely different directions, producing completely different endpoints and different midpoints along the way. This proves that a scalar (8 km) is insufficient to describe a journey — you need the full vector, with both magnitude AND direction, which is exactly what the driving question demands.

LESSON 7 (80 minutes): Translation Vector as a Transformation: Moving Shapes the Matatu Way	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To connect vectors to transformation geometry by establishing that a translation moves every point of a plane figure by the same vector, enabling learners to perform translations on a Cartesian plane and reverse-engineer the translation vector from an object and its image.
Knowledge	Learners know and understand that: (i) a translation vector describes both the magnitude and direction of a shift applied uniformly to every point of an object; (ii) the translation vector can be written in column vector form as (x over y) where x is the horizontal component and y is the vertical component; (iii) given an object and its image under translation, the translation vector is found by subtracting the position vector of any object point from the corresponding image point; (iv) translation is an isometric transformation — shape and size are preserved, only position changes.
Skills	Learners are able to: (i) perform a translation of a given plane figure on a Cartesian plane using a specified column translation vector; (ii) reverse-engineer the translation vector from an object and its image on a Cartesian plane; (iii) represent road-arrow instructions as column translation vectors; (iv) update and annotate the class cumulative geometric model on the Driving Question Board.
Attitudes	Learners appreciate the power of vectors as precise mathematical tools that fully describe movement — linking abstract notation to the lived reality of navigating Nairobi's roads — and demonstrate responsibility by working collaboratively and accurately on shared grid resources.
Key Inquiry Question	1. What is the meaning of the terms vector and scalar quantities? 2. Where are vectors used in real-life situations?
Purpose in Storyline	Lesson 7 is the EXPLAIN + TRANSLATION VECTOR AS TRANSFORMATION lesson in the eight-lesson evidence-gathering sequence. Having previously established what vectors are, how they are represented, added, and scaled, learners now deploy that knowledge to explain a new phenomenon layer: road directional arrows on Nairobi's streets are not mere signs — they are translation instructions that move every road user by the same vector. This lesson directly answers the driving question by showing that the reason the two matatu drivers end at different destinations is that their journeys are described by different translation vectors, not merely different distances. The lesson also bridges Sub-Strand 2.8 Vectors with the earlier Sub-Strands 2.1–2.3 (Similarity, Reflection, Rotation) by positioning translation as the fourth rigid transformation, completing learners' geometric toolkit.
Safety Notes	No physical hazards. Ensure squared-paper grids are large enough for accurate plotting. Remind learners to use sharp pencils and rulers for precise vector arrows to avoid measurement errors that could lead to misconceptions.

B. LESSON OVERVIEW

Learners begin by predicting how a one-way road arrow on a Nairobi street map could be expressed as a column vector. They then observe a worked demonstration of translating a triangle on a Cartesian plane using a given translation vector, noticing that every vertex shifts by exactly the same vector. In the Explain phase, pairs perform their own translations and reverse-engineer translation vectors from object–image pairs, connecting each result back to the matatu phenomenon. The Driving Question Board is updated with new consensus statements and remaining questions. Finally, learners revise their cumulative geometric model to incorporate translation vectors alongside earlier transformation diagrams. The road-safety context — directional arrows as translation instructions — runs throughout.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Intro to the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Translation of Vectors and Slope (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a simplified Nairobi street-grid handout (same map introduced in the anchoring phenomenon, now annotated with one-way road arrows for Tom Mboya Street going north and Kenyatta Avenue going west). Learners work individually for 3 minutes, then discuss in pairs for 2 minutes to answer: 'If a matatu at Kencom Bus Stop follows the Tom Mboya Street arrow for 3 blocks right and 2 blocks up, and another follows the Kenyatta Avenue arrow for 3 blocks left and 2 blocks up — can you write each journey as a column vector? Will they end up at the same place? Why or why not?' Learners record predictions in their exercise books using column vector notation they learned in earlier lessons.	Distribute the annotated Nairobi grid handout. Say: 'Look at the road arrows on your map. You already know vectors carry both size and direction. Today we ask: what happens when we apply a vector to an entire shape — every single point of it — all at once? Before I show you anything, predict.' Allow 3 minutes of individual silent thinking — WAIT TIME enforced, no prompting. Then say: 'Share your column vector with your partner. Convince each other.' After pair discussion, cold-call 3 pairs using a random name stick: ask each pair to read out their two column vectors and one-sentence prediction. Write all three pairs of vectors on the board without commenting on correctness. Say: 'Hold these predictions. We will come back to test them.'	Prediction Journaling with Column Vector Encoding — learners activate prior knowledge of column vectors and direction by encoding a familiar map context into mathematical notation before instruction. This surfaces misconceptions (e.g., thinking the same x-component means the same destination) and creates cognitive readiness for the observation phase.	Observable indicator: Can each learner independently write a column vector for a described journey? Teacher scans exercise books during the 3-minute individual phase, noting: (1) correct use of column vector brackets, (2) correct sign convention for direction. Flag any learner writing a scalar distance only — this indicates a persistent scalar-vs-vector misconception from earlier lessons that must be addressed before Phase 2.
Observe Phase 📹 **VIDEO:** [Intro to the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Translation of Vectors and Slope (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	The teacher projects (or draws on the board) a large Cartesian plane showing triangle KNB with vertices K(1,1), N(3,1), B(2,3) — named after Kencom, Nairobi, and Bustani to keep the Nairobi context alive. The teacher performs a live, step-by-step translation of every vertex by the translation vector t = (4 over 2), plotting K'(5,3), N'(7,3), B'(6,5) and drawing the image triangle K'N'B'. The teacher narrates each step aloud. Learners copy the diagram into their squared-paper books in real time, drawing arrows from each original vertex to its image. Learners then observe a second demonstration: a quadrilateral MJUA (Mombasa Junction Urban Arrow) is shown with its image M'J'U'A' already	At the board, say: 'I am going to move triangle KNB. Every single vertex — K, N, and B — will be shifted by the same translation vector t equals, column vector, 4 on top, 2 on the bottom. Watch what happens to K first.' Plot K(1,1), draw the vector arrow 4 right and 2 up, mark K'(5,3). Say: 'K moved 4 right and 2 up. Now N.' Repeat for N and B. Draw the full image triangle with a dashed outline. Say: 'What do you notice about the three arrows I drew?' WAIT 12 seconds. Cold-call 2 learners. Expect: 'They are all the same.' Confirm: 'Exactly — every point moved by the identical vector. That is the definition of a translation.' Then reveal the quadrilateral MJUA with its pre-	Parallel Deconstruction — learners first watch the teacher build a translation vertex-by-vertex (forward direction) then independently reverse-engineer the vector from a completed object–image pair (backward direction). Experiencing both directions of the same process deepens relational understanding: translation vector = image position vector minus object position vector.	Observable indicator: Do learners correctly compute the same column vector from all four vertices of MJUA? Teacher circulates and checks: (1) all four vectors are identical, (2) signs are correct, (3) brackets and notation are correct. A learner who gets different vectors for different vertices has not yet grasped the uniformity property — intervene immediately with: 'Pick vertex M only. How far right? How far up? Write that. Now check M only for J. Same?'

	plotted. Learners are challenged to measure the shift at each vertex and record the four column vectors they observe. They write what they notice about the four vectors.	drawn image. Say: 'Work out the translation vector from each vertex to its image. Write all four column vectors. What pattern do you see?' Allow 4 minutes. Cold-call 3 learners to read their vectors aloud. If a learner reads a wrong sign (e.g., negative instead of positive), say: 'Check your direction — which way did the x-component move? Left means negative.' Do not move on until consensus is reached that all four vectors are identical.		
Explain Phase 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Translation of Vectors and Slope (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs on a structured task sheet with three parts. PART A (Forward Translation): Translate quadrilateral GITI (representing a githeri-shaped irregular quad — a playful Kenya food reference) with vertices G(0,0), I(2,0), T(2,3), I_2(0,3) by translation vector p = (−3 over 4). Plot object and image on a Cartesian plane, label all vertices, and draw translation arrows. PART B (Reverse-Engineering): An object triangle SUK (Sukuma Wiki triangle) with vertices S(−4,1), U(−2,1), K(−3,4) and its image S'(1,3), U'(3,3), K'(2,6) is given. Learners must determine the translation vector. PART C (Road-Arrow Context): A section of Ngong Road is shown as a grid. Three road arrows are drawn at different points, all pointing in the same direction. Learners express each arrow as a translation vector in column form and explain in one sentence why all three arrows being equal vectors means any matatu following Ngong Road undergoes the same translation regardless of its starting position on that road.	Distribute task sheets and squared-paper books. Say: 'You have 18 minutes. Part A is forward — you apply the vector. Part B is backward — you find the vector. Part C connects it back to our matatu phenomenon. Work in your pairs.' During work time: (1) circulate to all pairs, spending no more than 90 seconds with any single pair; (2) if a pair is stuck on Part A negative x-component, prompt: 'Negative 3 in the x-component — which direction does that move you on the x-axis?'; (3) for Part B, if a learner subtracts in the wrong order, ask: 'You are trying to get FROM the object TO the image. So which point do you start with in your subtraction?'; (4) use a mini-whiteboard to record 2–3 interesting Part C responses anonymously for whole-class share. After 18 minutes, facilitate a 7-minute whole-class debrief. Ask: 'For Part B, what operation gave you the translation vector?' WAIT 10 seconds. Cold-call 3 pairs. Expect: 'Image minus object' or 'image position vector minus object position vector.' Write the formula on the board: t = position vector of	Bidirectional Problem Solving with Phenomenon Anchoring — by tackling both forward (apply vector) and reverse (find vector) problems and then explicitly mapping the result onto the Nairobi road context, learners consolidate the translation vector concept from multiple cognitive angles. Part C forces a return to the anchoring phenomenon, ensuring the mathematics is not disconnected from its explanatory purpose.	Observable indicators: (1) Part A — image vertices correctly plotted with all arrows identical in direction and length; (2) Part B — translation vector correctly computed as (5 over 2) using subtraction of position vectors; (3) Part C — learner's written sentence explicitly uses the word 'vector' and references both direction and magnitude. Collect task sheets at the end of the lesson for written feedback. Any learner who cannot complete Part B independently needs targeted support in Lesson 8 on position vectors and subtraction.

		image point − position vector of object point. Then show one anonymous Part C response and ask: 'How does this pair's answer help us explain why the two matatu drivers ended at different places even though they both drove 8 km?' WAIT 15 seconds. Cold-call 2 learners.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Translation of Vectors and Slope (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board (DQB) displayed at the front. The teacher reads aloud the driving question: 'Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?' Learners individually write two sticky-note responses on different coloured notes: (1) GREEN — a new 'We now know…' statement using today's learning about translation vectors; (2) ORANGE — a remaining or new question that today's lesson raised for them. Learners post their notes on the DQB under the appropriate columns. The teacher reads three green and two orange notes aloud and facilitates a 3-minute class consensus discussion to agree on one new consensus statement to add permanently to the DQB.	Point to the DQB and say: 'Look at what we have built since Lesson 1. Today we have added something powerful. Take 3 minutes — write your green note first: what do you now know that you did not know at the start of today? Connect it explicitly to the matatu phenomenon. Then write your orange note: what are you still wondering?' Allow 3 minutes silent writing — WAIT TIME enforced. Collect notes as learners post them, reading silently as they come in. Select three green notes that best capture today's SLO and one orange note that previews Lesson 8 content (e.g., about how vectors describe more complex journeys). Read each note aloud and say: 'Class, is this a fair statement of what we now know?' Build consensus. Write the agreed statement on the DQB anchor card: 'A translation vector moves every point of a shape by the same direction and magnitude — this is why the two matatu drivers ended at different places: they followed different translation vectors, not just different distances.'	Driving Question Board Consensus Protocol — structured sticky-note contributions followed by teacher-facilitated class consensus build both individual metacognitive reflection and collective knowledge documentation. The DQB serves as a running public record of how the class's understanding is growing across all eight lessons, making learning progress visible.	Observable indicator: Does each learner's green sticky note include (1) the phrase 'translation vector' or 'same vector', and (2) an explicit link to the matatu phenomenon or road arrows? A green note that only says 'translations move shapes' without referencing the phenomenon indicates surface-level engagement — note that learner for a follow-up question at the start of Lesson 8.
Model Building Phase 📹 **VIDEO:** [Intro to the	Learners retrieve their cumulative geometric model — a personal diagram they have been building across Lessons 1–6 showing the Nairobi street grid, vectors, and	Say: 'Open your cumulative model. Since Lesson 1 you have been building a diagram that explains the matatu phenomenon. Today you add the translation layer. You	Cumulative Model Revision — the running personal geometric model functions as an externalized thinking tool. Each revision forces learners to integrate new	Observable indicator: Does the learner's updated model include all three required elements — translated shape with equal vector arrows, connection to road arrows,

coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Translation of Vectors and Slope (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	their evolving understanding of the matatu phenomenon. Today they add a new layer: they draw a small Cartesian plane inset showing a triangle and its translated image connected by three identical translation vector arrows. They label this inset 'Translation = same vector applied to every point' and draw a connecting annotation arrow to the road-arrow section of their Nairobi map, writing: 'Road arrows on Tom Mboya Street are translation vectors.' They also add the formula t = image position vector − object position vector in a box. Learners who finish early add a second example using a quadrilateral and a different translation vector of their own choosing.	have 8 minutes. Your model must show: one, a translated shape with identical vector arrows; two, the connection to road arrows; three, the reverse-engineering formula.' Project a model template on the board showing the three required elements as labelled blank zones. Circulate and look for: (1) arrows that are not parallel and equal in length — redirect with 'Your translation vector is supposed to be the same for every vertex — check your arrows with a ruler'; (2) missing formula — prompt with 'How would a future learner use your model to find the translation vector if they only had the object and the image?'. With 2 minutes remaining, ask 2 learners to hold up their models and briefly narrate one thing they added today. Close with: 'In Lesson 8 we will pull everything together — all eight lessons — to fully answer our driving question. Your model today is your preparation for that final explanation.'	knowledge with prior understanding rather than treating each lesson as isolated. The physical act of annotating connections between the new translation layer and earlier vector representations reinforces coherence across the sub-strand.	and the reverse-engineering formula — in a coherent layout? A model that has the translated shape but no connection to the Nairobi road context shows compartmentalized thinking. A model that has only the formula with no diagram shows procedural over conceptual emphasis. Both are flagged for targeted feedback before Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully bridge from the abstract column vector to the physical translation of a shape — could they explain in their own words why every point moves by the same vector? (2) In Part B of the Explain phase, did the majority correctly compute the translation vector as image minus object, or did subtraction-direction errors persist? If more than 30% of learners reversed the subtraction, plan a 5-minute re-teach at the start of Lesson 8 using a concrete example: 'You start at Kencom (object) and walk to Westlands (image) — which position do you subtract from which to find your journey vector?' (3) Did the DQB green notes contain explicit phenomenon connections, or were they generic transformation statements? Generic notes suggest the phenomenon thread is weakening — re-anchor Lesson 8 explicitly with the matatu map from Lesson 1. (4) Were learners' cumulative models becoming richer and more connected across lessons, or are some learners treating each lesson's addition as a separate page? Consider pairing a strong model-builder with a struggling learner at the start of Lesson 8 for a 3-minute model-sharing protocol before the final synthesis activity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	On the Cartesian plane, when triangle KNB was translated by vector (4 over 2), every vertex — K, N, and B — shifted by exactly the same vector arrow, landing at K', N', B'. Similarly, all four vertices of quadrilateral MJUA shifted by the same vector. Road

	arrows on the Nairobi street-grid handout all pointed in the same direction and represented the same column vector.
What did I learn?	A translation moves every point of an object by the same translation vector, preserving shape and size but changing position. The translation vector can be found by subtracting the position vector of an object point from the position vector of its corresponding image point: t = image position vector − object position vector. Road directional arrows are real-world examples of translation vectors applied uniformly to all road users.
How does this explain the phenomenon?	The two matatu drivers both travelled 8 km from Kencom Bus Stop but ended at different destinations because they followed different translation vectors — not just different distances. One driver's journey vector pointed north-east toward Westlands; the other pointed south toward South B. A scalar (distance = 8 km) cannot distinguish these journeys. Only a vector — capturing both magnitude and direction — completely describes where you will end up. Translation as a transformation makes this mathematically precise: every point of the matatu (or any object) is shifted by the same vector, which is why direction is the deciding factor in the final destination.

LESSON 8 (80 minutes): Model Build, Consolidation & Final Explanation — Vectors in Nakuru: Synthesising the Complete Journey

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate all vector concepts through a group multi-leg journey problem set in Nakuru, and to produce a final written explanation that fully answers the unit driving question.
Knowledge	Learners will demonstrate that: (i) a vector encodes both magnitude and direction; (ii) the resultant of a multi-leg vector journey is found by adding column vectors; (iii) the magnitude of a vector is computed using Pythagoras' theorem; (iv) the midpoint of a vector is the average of the position vectors of its endpoints; and (v) a translation vector maps a start point to an endpoint and is itself a vector quantity.
Skills	Learners will be able to: represent a multi-leg journey as a sequence of column vectors on a sketch map of Nakuru; compute the resultant displacement vector by summing column vectors; find the magnitude of the resultant vector; identify the midpoint of the overall journey vector; state the translation vector that maps the start point to the endpoint; and present their vector model to peers with justification.
Attitudes	Learners appreciate the power and precision of vectors in describing real-world journeys, value collaborative mathematical reasoning, and take pride in constructing a complete and coherent vector model that resolves the anchoring phenomenon.
Key Inquiry Question	Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?
Purpose in Storyline	This is the final lesson of the Vectors unit. Learners synthesise every concept introduced in Lessons 1–7 (scalar vs vector, notation, representation, equivalence, addition, scalar multiplication, column vectors, position vectors, magnitude, midpoint, and translation) in a single rich group task centred on a Nakuru town sketch map. By producing a full vector model of a multi-leg journey and then writing a personal final explanation of the driving question, learners complete the epistemic arc that began with the puzzling Nairobi matatu phenomenon: distance alone cannot locate you — you need a vector.
Safety Notes	No physical hazards. Ensure all printed sketch maps and rulers are shared equitably. If the lesson uses digital devices to display the Nakuru map, ensure screen brightness is comfortable and devices are handled responsibly.

B. LESSON OVERVIEW

In this 80-minute capstone lesson, learners work in groups of four to solve a structured multi-leg vector journey problem set on a simplified sketch map of Nakuru town (landmarks include Nakuru CBD matatu terminus, Westside Mall, Nakuru National Park gate, Lake Nakuru viewpoint, and the Agricultural Society of Kenya [ASK] Showground). Each group receives the map, a task card, and a grid overlay. They describe each leg as a column vector, sum the vectors to find the resultant displacement, compute its magnitude, find the midpoint of the journey, and state the translation vector from start to finish. Groups then present their vector models to the class. The lesson closes with a Final DQB Review in which every learner independently writes a full answer to the driving question, explicitly comparing their Day-1 thinking with their current understanding. The teacher facilitates a whole-class share-out of the final explanation and celebrates the intellectual journey the class has made from scalar distance to vectors.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the original Nairobi matatu phenomenon map (Kencom → Westlands vs Kencom → South B, both 8 km) projected or printed. They are asked, individually and silently for 3 minutes, to write in their exercise books: 'Without looking at your notes, write everything a vector must tell us that a scalar distance cannot — and predict whether you could fully describe a three-leg matatu journey in Nakuru using only numbers without direction.' They then share predictions with their elbow partner for 2 minutes. Three pairs are cold-called to share with the class.	Re-display the original phenomenon map and say: 'We started this unit with two matatu drivers, both travelling 8 km from Kencom, ending up in completely different places. Today is the day we prove — with mathematics — exactly why that happens and build the complete tool that fixes it.' Distribute the individual predict task. Set a visible 3-minute timer. Enforce silent individual thinking. After 3 minutes say: 'Turn to your elbow partner. You have 2 minutes — compare your lists.' After 2 minutes, cold-call exactly 3 pairs using a random name stick: 'Tell us one thing a vector gives us that a scalar cannot.' Wait 12 seconds after each prompt before accepting responses. Record key phrases (direction, sense, resultant, translation) on the board under the heading 'What we already know.' Do NOT correct or add to this list yet — it serves as the pre-assessment baseline.	Activating Prior Knowledge / Phenomena Reconnection — By returning to the original phenomenon before the final build, learners retrieve and consolidate their conceptual schema. The contrast between their Day-1 prediction (recorded in Lesson 1) and their current fluent response makes their own learning growth visible and motivates the synthesis task ahead.	Observable indicator: each learner produces a written list that includes at minimum 'direction' and 'magnitude' without prompting. The teacher scans 6–8 books during the pair-share to check for any persistent misconceptions (e.g., conflating displacement with distance). Any misconception heard during cold-call is noted on a sticky note for targeted follow-up during the group task.
Observe Phase 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive: (1) a printed A3 simplified sketch map of Nakuru town with a square grid overlay (each grid square = 1 km), compass rose, and five labelled landmarks; (2) a Task Card describing a three-leg matatu/boda-boda journey: Leg 1 — from Nakuru CBD Terminus, travel 3 km East and 4 km North to Westside Mall; Leg 2 — from Westside Mall, travel 5 km East and 2 km South to ASK Showground; Leg 3 — from ASK Showground, travel 2 km West and 6 km North to Lake Nakuru Viewpoint. Learners read the task card carefully, identify start and	Form groups of four by counting off (1–2–3–4 around the room) to ensure mixed ability. Distribute maps and task cards. Say: 'Your job as a group has five parts, written on your task card. Read the task card in full before you touch a pencil — you have 2 minutes to read silently.' After 2 minutes: 'Now trace each leg on your map using a different colour for each leg. Label each arrow. You are detectives — the map is your evidence.' Circulate and look for: correct directional arrows; correct grid counting; groups who have labelled start and end of each leg. After 5 minutes of map tracing, ask	Evidence Gathering via Concrete Representation — The physical act of tracing directional legs on a real Kenyan town map anchors the abstract column vector operations in a tangible spatial context. Colour-coding each leg visually separates the components, making the additive structure of vector journeys perceptually salient before any algebra is written.	Observable indicator: all groups have correctly traced three directed arrows on the map with start/end labels and are able to identify that the straight-line distance from start (Nakuru CBD) to finish (Lake Nakuru Viewpoint) is shorter than the total path distance. Teacher notes any group that draws undirected line segments (a sign that direction is not yet encoded) and targets that group first in Phase 3.

	end points on the map, trace each leg with a coloured pen, and annotate each arrow with the leg label. They observe the path and note: 'The total distance walked is the sum of all leg lengths. But where do we actually end up — and how far in a straight line from where we started?'	the whole class: 'Has every group located where the journey ends? Hold up your map.' Visually confirm all groups have the correct endpoint. Do NOT give the answer — simply confirm with a thumbs up or redirect groups that are off. Say: 'Notice — the total path distance and the straight-line distance from start to finish are NOT the same. Your mathematics today will tell us exactly how far and in what direction you have travelled overall.'		
Explain Phase 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	Each group now works through the five mathematical tasks on their Task Card, recording all working on a large sheet of manila paper (for gallery display): Task 1 — Write each leg as a column vector. Task 2 — Add the three column vectors to find the resultant displacement vector. Task 3 — Find the magnitude of the resultant displacement vector (to 2 decimal places). Task 4 — Given that the start point has position vector relative to a fixed origin O at the bottom-left of the grid, find the position vector of the endpoint and hence determine the midpoint of the journey (start to finish). Task 5 — State the translation vector that maps the start point to the endpoint and explain in one sentence what this vector tells a matatu driver. After completing calculations, each group writes a 'Vector Story' of three sentences summarising the journey using correct mathematical language. Groups then do a 10-minute Gallery Walk: half the groups stand at their poster while the other half rotate and leave one written question on a sticky note at each poster. Groups return to	Distribute large manila paper and markers. Say: 'All working must be visible on the manila paper — not just answers. A mathematician shows their reasoning, not just their result.' Set a 20-minute timer for the five tasks. Circulate using targeted questioning: For Task 1: 'How do you write 3 km East and 4 km North as a single column vector? What is the sign convention you are using?' Wait 12 seconds. For Task 2: 'When you add these column vectors, what does the result represent geographically? Point to it on the map.' Wait 12 seconds. For Task 3: 'Which theorem connects the two components of a vector to its magnitude? Why does that theorem apply here?' Wait 10 seconds. For Task 4: 'What is the formula for the midpoint vector? What does that point represent on your Nakuru map?' For Task 5: 'If I told a boda-boda rider only the magnitude of the resultant — would they know where to go? What is missing?' After 20 minutes, initiate the Gallery Walk. Say: 'Group members 1 and 2 stay at your poster; members 3 and 4 rotate clockwise. You have 4	Collaborative Modelling with Gallery Walk Critique — The large-format group poster makes mathematical reasoning public and critiqueable. The Gallery Walk activates peer-assessment: learners must read and evaluate another group's vector model, which requires them to reconstruct the reasoning themselves. Responding to sticky-note questions deepens metacognitive engagement. The 'Vector Story' bridges symbolic mathematics back to the geographical narrative, reinforcing that vectors describe real journeys.	Observable indicators: (1) Column vectors are written with correct bracket notation and correct signs for direction. (2) Resultant vector is obtained by adding corresponding components (not magnitudes). (3) Magnitude is computed as $\sqrt{(\Delta x^2 + \Delta y^2)}$. (4) Midpoint vector is computed as ½(position vector of start + position vector of end). (5) Translation vector equals the resultant displacement vector. Teacher uses a 3-point checklist (correct notation / correct operation / correct interpretation) on a clipboard to mark each group's poster during the gallery walk. Any group scoring 1/3 on notation receives a 2-minute teacher conference before Phase 4.

	answer the sticky-note question.	minutes at each poster. Leave one genuine mathematical question on a sticky note.' After rotations, all members return: 'Read the sticky-note question left at your poster. Discuss it as a group and write your answer directly on the manila paper.' Cold-call 2 groups to share the question they received and their answer.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The class reassembles. The teacher reveals the full DQB that has been built across all 8 lessons. Learners silently read every question card posted since Lesson 1 (approximately 2 minutes). Each learner then individually writes their Final DQB Answer in their exercise book: a structured paragraph (minimum 8 sentences) answering the driving question — 'Why do two matatu journeys of exactly the same distance from Nairobi's CBD end up at completely different destinations — and what mathematical tool do we need to describe a journey completely?' The answer must include: (a) the definition of a scalar and why distance alone is insufficient; (b) the definition of a vector and what it adds; (c) at least one column vector example from the Nakuru task; (d) how the resultant displacement was found; (e) the role of translation vectors; (f) a direct reference back to the Kencom → Westlands vs Kencom → South B phenomenon. Learners who finish early add a reflection: 'What was the most surprising thing vectors can do that I did not expect on Day 1?'	Re-project or physically point to the full DQB. Say: 'Look at every question we have asked since Lesson 1. Some of these questions puzzled us. Some we answered along the way. Today, we answer the big one — together and individually.' Give 2 minutes of silent reading of the DQB. Then say: 'Now — in your exercise book — write your Final DQB Answer. This is your personal proof that you understand vectors. It must be in your own words. You have 12 minutes.' Set a visible timer. Enforce silent individual writing. After 12 minutes, cold-call 4 learners (two high-confidence, two who showed earlier misconceptions) to read one sentence from their answer each. After each reading, ask the class: 'Does everyone agree with this sentence? Give a thumbs up, thumbs sideways, or thumbs down.' Address any thumbs-down responses immediately with: 'Tell me — what would you change and why?' Conclude this phase by pointing to the original phenomenon map and saying: 'The matatu drivers were not doing anything wrong. They both travelled 8 km. But one went North-West, the other went South. Same scalar. Different vectors. That is the entire lesson.'	Written Final Explanation / DQB Closure — The individual written answer is the highest-stakes sensemaking move in the unit. It requires learners to integrate all conceptual threads into a coherent causal narrative. By structuring the answer around the original phenomenon, learners experience closure: the puzzle that opened the unit is now fully resolved by the mathematics they have built. The cold-call share-out and thumbs assessment creates a public accountability structure that surfaces any residual confusion before the unit ends.	Observable indicator: learner final DQB answers include all six required elements (a–f listed in the learner experience). Teacher collects exercise books at the end of the lesson and uses a 6-point checklist (one point per element) to score each answer. Scores of 4+ indicate meeting expectations. Scores of 3 or below trigger a follow-up consolidation task in the next available lesson. Teacher also checks that learners explicitly use the word 'vector' (not just 'direction') and reference the column vector notation.

Model Building Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners open to their Lesson 1 initial model (drawn in the first lesson of the unit — typically an arrow or a simple distance label on a line). They draw their Final Model beside it on a fresh page or on the back of the page. The Final Model must include: (i) a vector arrow with clear direction and magnitude labelled; (ii) component column vector notation; (iii) resultant vector notation; (iv) the translation vector mapping start to end; (v) a note distinguishing scalar (distance) from vector (displacement). Below both models, learners write two sentences: 'My model changed because...' and 'The most important addition to my model is... because...' Three volunteers share their model comparison with the class using the document camera or by holding up their book. The class applauds each presenter. Teacher closes by saying: 'Every one of you arrived here thinking distance was enough to describe a journey. You leave knowing that a vector — magnitude AND direction — is the complete description. That is what a mathematician does: they find the precise tool for the job.'	Say: 'Turn to your very first lesson in this unit. Find the model or drawing you made on Day 1 about how to describe the matatu journey. You have 1 minute to find it.' Allow 1 minute. Then: 'Now draw your Final Model on the next blank page. Use everything you know. Show column vectors, resultant, translation. You have 8 minutes.' Set timer. Circulate, looking specifically for: direction arrows clearly labelled; column vector notation correctly used; distinction between scalar and vector explicitly stated in the written sentences. After 8 minutes, invite 3 volunteers — try to include at least one learner who struggled earlier in the unit. Use the document camera if available, or ask learners to hold up their books and describe verbally. After each share, prompt the class: 'What is one thing in this final model that was NOT in the Day-1 model?' Cold-call 2 responses per presenter (6 total). Close the lesson: 'The DQB is complete. The models are complete. Vectors are not just symbols on paper — they are the mathematics of direction, of journeys, of GPS, of the engineers who designed the roads between Nakuru and Nairobi. You now have that tool.' Assign the class activity (described in Teacher Reflection) as consolidation.	Model Revision / Before-and-After Comparison — Placing the initial and final models side by side makes conceptual change visible and explicit. This metacognitive strategy (often called 'model evolution tracking') allows learners to see their own intellectual growth as evidence, reinforcing the value of the inquiry process. The verbal sharing of model comparisons creates a collective celebration of learning that also serves as a final informal assessment of the unit's core conceptual goal.	Observable indicators: (1) Final model contains all five required elements. (2) Written 'My model changed because...' sentence correctly attributes the change to the addition of direction/vector concept. (3) Distinction between scalar and vector is explicit and accurate. Teacher uses the 3-group checklist (Exceeds / Meets / Approaches Expectations, aligned to the KICD rubric for 'Ability to represent vectors geometrically, perform operations on vectors, and determine the magnitude of vectors') to informally grade the final model. Results inform the teacher's report on sub-strand 2.8 attainment.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) DQB Final Answers — Did learners' written answers demonstrate a complete and accurate resolution of the driving question, or did any answers still conflate distance with displacement? Collect and score all exercise books using the 6-point checklist before the next lesson. (2) Group Task Performance — Which groups struggled most with Task 3 (magnitude) or Task 4 (midpoint position vector)? These are the two most computationally demanding tasks and may require a short consolidation class activity in the first available lesson of the next sub-strand period. (3) Model Comparison Quality — Did the majority of final models include all five required elements? If column vector notation was absent or incorrect in more than 30% of final models, plan a 10-minute warm-up activity at the start of the next Mathematics lesson to revisit notation conventions. (4) Equity of Voice — Were learners who struggled earlier in the unit (identified in Lessons 2–5) able to contribute meaningfully to both the group task and the DQB final answer? If not, consider a differentiated consolidation worksheet with scaffolded column vector tasks set in familiar Kenyan food market contexts (e.g., a vendor walking between stalls at Gikomba Market). (5) SUGGESTED CLASS ACTIVITY (to assign as consolidation or homework): 'A farmer in Kericho walks from her tea farm to the cooperative collection point. She walks 6 km East then 8 km North. (a) Express her journey as a column vector. (b) Find the magnitude of her displacement. (c) Find the midpoint of her displacement vector if her start position vector from the cooperative office is (2, 1). (d) What translation vector maps the start point to the collection point? (e) Would knowing only that she walked 10 km tell us where she ended up? Explain using the terms scalar and vector.' This activity assesses all five key skills of the sub-strand in a single Kenyan-contextualised problem.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners traced a three-leg boda-boda/matatu journey on a sketch map of Nakuru town (CBD Terminus → Westside Mall → ASK Showground → Lake Nakuru Viewpoint), colour-coding each leg and noting that the total path distance and the straight-line displacement from start to finish are not the same quantity.
What did I learn?	Learners confirmed that each leg of a journey can be expressed as a column vector capturing both magnitude and direction; that column vectors are added component-wise to give the resultant displacement vector; that the magnitude of the resultant is found using Pythagoras' theorem; that the midpoint vector is the average of the start and end position vectors; and that the translation vector mapping start to finish is identical to the resultant displacement vector — together proving that scalar distance alone cannot locate a destination.
How does this explain the phenomenon?	Learners wrote a Final DQB Answer resolving the anchoring phenomenon: the two matatu drivers from Kencom both travelled 8 km (equal scalars) but in different directions, producing different displacement vectors and therefore entirely different endpoints. A vector — encoding both magnitude and direction in column form — is the precise mathematical tool needed to describe a journey completely, predict a destination, and define the translation that maps one location to another.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.